STULZ

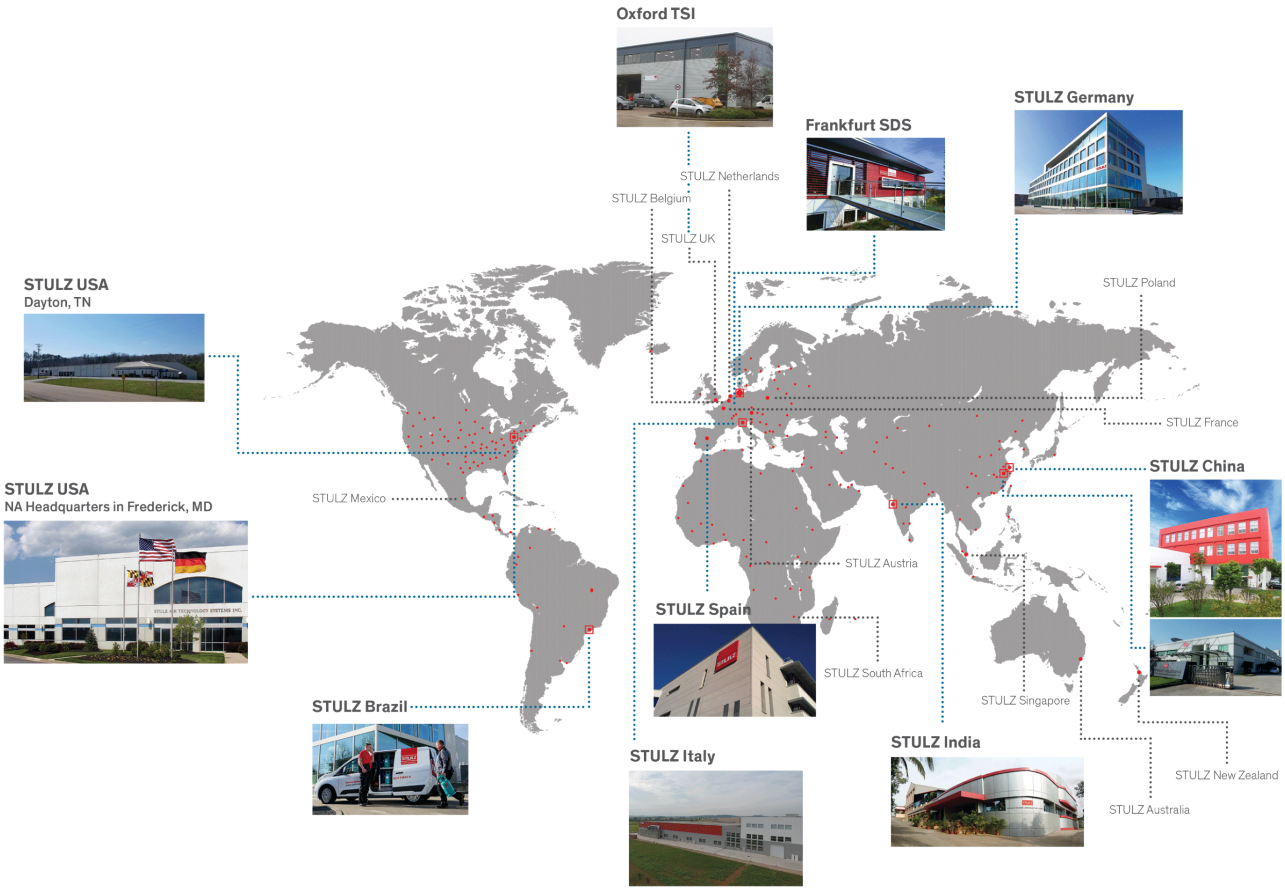
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CyberAir® DX

Engineering Manual

Perimeter Mounted Precision Air Conditioners
21 kW - 105 kW / 60 Hz



ABOUT STULZ

Since it was founded in 1947, the STULZ company has evolved into one of the world's leading system suppliers of computer room air conditioners and air handlers.

STULZ has seen continual international expansion of its precision cooling business, specialising in temperature and humidity control for data centers and telecommunications installations.

STULZ has ten production plants (2 in Germany, 2 in the USA, Italy, Great Britain, Spain, 2 in China, Brazil and India) and twenty subsidiaries (in Germany, USA, France, Italy, Great Britain, Belgium, Brazil, the Netherlands, New Zealand, Mexico, Austria, Poland, Spain, Singapore, China, India, Indonesia, South Africa, Sweden, and Australia).

STULZ works with sales and service partners in over 140 other countries, and therefore boasts an international network of cooling specialists.

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The benchmark of flexibility and energy efficiency

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Model Nomenclature								
CFX-XXX-XXX-X-XX								
Model	**Air Pattern**	**Capacity (kW)**	**Cooling Method**	**Number of Circuits**	**Heat Rejection**	**Secondary Cooling Options**	**Fan Location**	**Additional Special Options**
CyberAir Floor Mounted Solution	D = Downflow U = Upflow R = Rear Return	021 028 035 042 053 070 088 105	D = DX	2 = Dual Circuit	A = Air G = Glycol W= Water	0 = None F = Water-Side Economizer W= Alternate Water Source	I = In Unit	0 = Standard T = Tandem Compressors S = Special
CF	**D**	**105**	**D**	**2**	**W**	**0**	**I**	**0**

Example:105 kW Capacity, Dual Circuit, Water Cooled, Direct Expansion, Downflow, Floor Mounted System; with Internal EC Fans, and No Additional Options: **CFD-105-D2W-0-I0**

Selected Features

Downflow / Upflow Models 021 - 105

Models	021 thru 105
Temperature Control	
2-Stage Cooling Mode	Standard
4-Stage Cooling Mode (Tandem Compressors)	Optional (Models CFD/U-053 thru CFD/U-105)
Electric Reheating	Standard
Cooling Only (No Humidity Control)	Optional
Hot Water Reheat	Optional
Hot Gas Reheat	Optional
SCR Fired Electric Reheating	Optional
Humidity Control	
Electrode Canister Steam Humidifier	Standard
Dehumidification Mode - Electric Reheat	Standard
Hot Gas Reheat	Optional
Controls	
E² Color Touchscreen Graphic Microprocessor with Alarms	Standard
Remote BMS Communications	Optional
Cabinet	
Galvannealed Steel, Powder Coated Finish	Standard
Stainless Steel Condensate Drain Pan	Standard
½" Thick, 2 lb/ft² Density Thermal & Sound Insulation	Standard
Floor Stand (Adjustable)	Optional
Filters / Plenums	
2 or 3-Way Plenum Box (Up-Flow Units)	Optional
Return Air Plenum Extension (Down-Flow Units)	Optional
4", MERV8 Rated Pleated Filters	Standard
4", MERV11 Rated Pleated Filters	Optional
DX Refrigeration Circuit	
Evaporator Side Feed Hot Gas Bypass	Optional (not available on models with Tandem Compressors)
R410A Refrigerant	Standard
Dual Scroll Compressors w/ Rotalock Service Valves (N/A with Tandem)	Standard
Tandem Scroll Compressors	Optional (Models CFD/U-053 thru CFD/U-105)
High Efficiency Aluminum Fin / Copper Tube Coils	Standard

Electronic Expansion Valves	Standard
Refrigerant Sight Glass & Filter/ Drier Strainer	Standard
Blowers / Motors	
Backward Inclined, Plenum Style Fan, with an EC Motor and high efficient composite impeller	Standard
Electrical	
3-Phase Power Supply	See Electrical Table
Multi-Voltage Control Transformer (24V Class 2)	Standard
Modular Motor Controllers with Integral Circuit Breakers	Standard
Safety Features	
Audible/ Visual Local & Remote Alarms	Standard
Main Power Circuit Breaker Unit Mounted	Standard
High/ Low Refrigerant Pressure Switches	Standard
Motor Overload Protection	Standard
Specific Model Standard Features:	
Air Cooled	
Remote Outdoor, Air Cooled Condenser rated for -20°F to 95°F Ambient	Standard
Low Ambient Control to -30°F and High Ambient to 105°F	Optional
Water / Glycol Cooled	
2-Way, 600 psig Water / Glycol Regulating Valves	Standard
3-Way, 600 psig Water / Glycol Regulating Valves	Optional
Free Cooling / Alternate Water Source Systems	
2-Way, 600 psig Water/Glycol Control Valve	Standard (AWS)
3-Way, 400 (042-105 kW systems) or 600(021-035 kW systems) psig Water/Glycol Control Valve	Standard (FC); Optional (AWS)
Code Conformance	
NRTL Conformance Compliance to UL 1995 Standard	Standard

Performance / Capacity Data

Air Cooled Systems – Downflow / Upflow Models 021 – 042

Models:	CFD-021-D2A	CFU-021-D2A	CFD-028-D2A	CFU-028-D2A	CFD-035-D2A	CFU-035-D2A	CFD-042-D2A	CFU-042-D2A
Evaporator Blower / Motor - Backward Inclined Direct Driven EC								
Nominal Horsepower per Blower, hp	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.7
Rated Air Flow, CFM @ .2 in. H_2O ESP	6,000	6,000	6,000	6,000	6,000	6,000	8,000	8,000
Quantity of Blowers	1	1	1	1	1	1	2	2
Evaporator Coil - Aluminum Fin, Copper Tube								
Direct Expansion (DX) Coil								
Rows / (Face Area, ft^2)	2 (21.6)	2 (21.6)	2 (21.6)	2 (21.6)	2 (21.6)	2 (21.6)	3 (24.7)	3 / (24.7)
Face Velocity, fpm	278	278	278	278	278	278	323	323
Alternate Water Source (AWS) Coil (Optional)								
Coil Rows (Face Area, ft^2)	4 (21.6)	4 (21.6)	4 (21.6)	4 (21.6)	4 (21.6)	4 (21.6)	4 (24.7)	4 / (24.7)
Face Velocity, fpm	278	278	278	278	278	278	323	323
CV (AWS valve, 3-way 400 PSI)	11	11	11	11	11	11	20	20
Compressors - R-410A								
Type (Qty)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll / (2)
Input per Compressor, W	3,900	3,900	4,710	4,710	5,340	5,340	5,880	5,880
Total Heat of Rejection, kW (MBH)	38.1 (130)	38.1 (130)	44.1 (151)	44.1 (151)	48.3 (165)	48.3 (165)	58.1 (199)	58.1 (199)
Reheat - Performance Capacities DO NOT Include Evaporator Motor Heat @ Rated CFM & ESP								
Electric Reheat - Finned Tubular								
Number of Stages	1	1	1	1	1	1	2	2
Total Nominal Capacity, kW (MBH)	9.0 (30)	9.0 (30)	9.0 (30)	9.0 (30)	9.0 (30)	9.0 (30)	18.0 (61)	18.0 (61)
Hot Gas Reheat - with 3-Way Heat Reclaim Valve								
Total Capacity, kW (MBH)	3.9 (13)	3.9 (13)	4.7 (16)	4.7 (16)	5.3 (18)	5.3 (18)	5.8 (20)	5.8 (20)
Hot Water Reheat - Rated at 180°F EWT / 160°F LWT & 72°FDB/60.0°FWB EAT								
Total Capacity, kW (MBH)	19.4 (66)	23.8 (81)	19.4 (66)	23.8 (81)	19.4 (66)	23.8 (81)	24.2 (82)	29.8 (101)
Flow Rate, GPM (Pressure Drop, ft H_2O)	7 (4)	9 (5)	7 (4)	9 (5)	7 (4)	9 (5)	9 (5)	11 (8)
Control Valve	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating
Steam Reheat - Rated at 5 psig & 72°FDB EAT								
Total Capacity, kW (MBH)	44.5 (152)	61.9 (211)	44.5 (152)	61.9 (211)	44.5 (152)	61.9 (211)	58.2 (198)	79.2 (270)
Condensate, lbs/hr	158	220	158	220	158	220	207	281
Control Valve	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating
Humidification - Electrode Steam Canister with Adjustable Output								
Steam Output, lbs/hr (Power Input, kW)	4-15 (5.1)	4-15 (5.1)	4-15 (5.1)	4-15 (5.1)	4-15 (5.1)	4-15 (5.1)	4-15 (5.1)	4-15 (5.1)
Control Type	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating
Filters - Disposable, MERV 8 Average Dust Spot Efficiency (Standard), MERV 11 Average Dust Spot Efficiency (Optional)								
Actual Size, H in. x W in. x D in. (Qty)	37.0 x 18.5 x 3.75 (3)	35.9 x 24.8 x 3.8 (1) 35.9 x 18.9 x 3.8 (1) 21.0 x 18.9 x 3.8 (1)	37.0 x 18.5 x 3.8 (3)	35.9 x 24.8 x 3.8 (1) 35.9 x 18.9 x 3.8 (1) 21.0 x 18.9 x 3.8 (1)	37.0 x 18.5 x 3.8 (3)	35.9 x 24.8 x 3.8 (1) 35.9 x 18.9 x 3.8 (1) 21.0 x 18.9 x 3.8 (1)	37.0 x 21.0 x 3.8 (3)	35.9 x 24.8 x 3.8 (2) 24.8 x 21.0 x 3.8 (1)
Connection Sizes - Copper								
Hot Gas Line, in. OD (2 per Unit)	5/8	5/8	5/8	5/8	5/8	5/8	7/8	7/8
Liquid Line, in. OD (2 per Unit)	1/2	1/2	1/2	1/2	1/2	1/2	5/8	5/8
AWS In & Out, OD in.	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	2 1/8	2 1/8
Condensate Drain without Pump, OD in.	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8
Condensate Drain with Pump, OD in.	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2
Humidifier Inlet, OD in.	1/4	1/4	1/4	1/4	1/4	1/4	1/4	1/4
Hot Water Reheat In & Out, OD in.	7/8	7/8	7/8	7/8	7/8	7/8	7/8	7/8
Physical Data								
Approximate Unit Weight, lbs	1,750	1,950	1,750	1,950	1,750	1,950	2,200	2,400
Approximate Shipping Weight, lbs	1,850	2,050	1,850	2,050	1,850	2,050	2,350	2,550
Unit Dimensions, H in. x W in. x D in.	76.2 x 77.4 x 40.3	88.1 x 77.4 x 40.3 (91.1 x 77.4 x 40.3 w/ Hot Gas/Water Reheat)	76.2 x 77.4 x 40.3	88.1x 77.4 x 40.3 (91.1x 77.4 x 40.3 w/ Hot Gas/Water Reheat)	76.2 x 77.4 x 40.3	88.1 x 77.4 x 40.3 (91.1 x 77.4 x 40.3 w/ Hot Gas/Water (Reheat)	76.2 x 89.1 x 40.3	88.1 x 88.2 x 40.3 (91.1 x 88.2 x 40.3 w/ Hot Gas/Water Reheat)
Shipping Dimensions, H in. x W in. x D in.	83.0 x 80.0 x 46.0	95.0 x 80.0 x 46.0 (98.0 x 80.0 x 46.0 w/ Hot Gas/Water Reheat)	83.0 x 80.0 x 46.0	95.0 x 80.0 x 46.0 (98.0 x 80.0 x 46.0 w/ Hot Gas/Water Reheat)	83.0 x 80.0 x 46.0	95.0 x 80.0 x 46.0 (98.0 x 80.0 x 46.0 w/ Hot Gas/Water Reheat)	83.0 x 95.0 x 46.0	95.0 x 95.0 x 46.0 (98.0 x 95.0 x 46.0 w/ Hot Gas/Water Reheat)

Performance / Capacity Data

Air Cooled Systems – Downflow / Upflow Models 053 – 105

Models	CFD-053-D2A	CFU-053-D2A	CFD-070-D2A	CFU-070-D2A	CFD-088-D2A	CFU-088-D2A	CFD-105-D2A	CFU-105-D2A
Evaporator Blower / Motor - Backward Inclined Direct Driven EC								
Nominal Horsepower per Blower, hp	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.7
Rated Air Flow, CFM @ 0.2 in. H_2O ESP	9,000	9,000	10,000	10,000	15,750	15,750	15,750	15,750
Quantity of Blowers	2	2	2	2	3	3	3	3
Evaporator Coil - Aluminum Fin, Copper Tube								
Direct Expansion (DX) Coil								
Rows (Face Area, ft^2)	3 (24.7)	3 (24.7)	3 (24.7)	3 (24.7)	3 (39.8)	3 (39.8)	3 (39.8)	3 (39.8)
Face Velocity, fpm	364	364	404	404	395	395	395	395
Alternate Water Source (AWS) Coil								
Coil Rows (Face Area, ft^2)	4 (24.7)	4 (24.7)	4 (24.7)	4 (24.7)	4 (39.8)	4 (39.8)	4 (39.8)	4 (39.8)
Face Velocity, fpm	364	364	404	404	395	395	395	395
CV (AWS valve, 3-way 400 PSI)	20	20	20	20	28	28	28	28
Compressors - R-410A								
Type (Qty)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)
Input per Compressor, W	7,900	7,900	10,390	10,390	13,320	13,320	15,640	15,640
Total Heat of Rejection, kW (MBH)	73.7 (252)	73.7 (252)	92.7 (317)	92.7 (317)	125.7 (429)	125.7 (429)	139.8 (477)	139.8 (477)
Reheat - Performance Capacities Do Not Include Evaporator Motor Heat @ Rated CFM & ESP								
Electric Reheat - Finned Tubular								
Number of Stages	2	2	2	2	3	3	3	3
Total Nominal Capacity, kW (MBH)	18.0 (61)	18.0 (61)	18.0 (61)	18.0 (61)	27.0 (92)	27.0 (92)	27.0 (92)	27.0 (92)
Hot Gas Reheat - with 3-Way Heat Reclaim Valve								
Total Capacity, kW (MBH)	7.9 (26)	7.9 (26)	10.3 (35)	10.3 (35)	13.3 (45)	13.3 (45)	15.6 (53)	15.6 (53)
Hot Water Reheat - Rated at 180°F EWT / 160°F LWT & 72°FDB/60.0°FWB EAT								
Total Capacity, kW (MBH)	25.5 (87)	31.1 (106)	26.6 (91)	32.2 (110)	45 (153)	56.1 (191)	45 (153)	56.1 (191)
Flow Rate, GPM / (Pressure Drop, ft H_2O)	9 / (6)	11 / (9)	10 / (6)	12 / (9)	16 / (18)	20 / (27)	16 / (18)	20 / (27)
Control Valve	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating
Steam Reheat - Rated at 5 psig & 72°FDB EAT								
Total Capacity, kW (MBH)	62.8 (214)	84.7 (289)	66.8 (228)	89.1 (304)	105.2 (359)	146.8 (501)	105.2 (359)	146.8 (501)
Condensate, lbs/hr	223	301	237	316	374	521	374	521
Control Valve	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating
Humidification - Electrode Steam Canister with Adjustable Output								
Steam Output, lbs/hr (Power Input, kW)	4-15 (5.1)	4-15 (5.1)	4-15 (5.1)	4-15 (5.1)	4-25 (8.5)	4-25 (8.5)	4-25 (8.5)	4-25 (8.5)
Control Type	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating
Filters - Disposable, MERV 8 Average Dust Spot Efficiency (Standard), MERV 11 Average Dust Spot Efficiency (Optional)								
Actual Size, H in. x W in. x D in. (Qty)	37.0 x 21.0 x 3.8 (3)	35.9 x 24.8 x 3.8 (2) 24.8 x 21.0 x 3.8 (1)	37.0 x 21.0 x 3.8 (3)	35.9 x 24.8 x 3.8 (2) 24.8 x 21.0 x 3.8 (1)	37.0 x 18.5 x 3.8 (5)	35.9 x 24.8 x 3.8 (3) 24.8 x 21.0 x 3.8 (2)	37.0 x 18.5 x 3.8 (5)	35.9 x 24.8 x 3.8 (3) 24.8 x 21.0 x 3.8 (2)
Connection Sizes - Copper								
Hot Gas Line, in. OD (2 per Unit)	7/8	7/8	7/8	7/8	1 3/8	1 3/8	1 3/8	1 3/8
Liquid Line, in. OD (2 per Unit)	5/8	5/8	5/8	5/8	7/8	7/8	7/8	7/8
AWS In & Out, OD in.	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8
Condensate Drain without Pump, OD in.	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8
Condensate Drain with Pump, OD in.	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2
Humidifier Inlet, OD in.	1/4	1/4	1/4	1/4	1/4	1/4	1/4	1/4
Hot Water Reheat In & Out, OD in.	7/8	7/8	7/8	7/8	7/8	7/8	7/8	7/8
Physical Data								
Approximate Unit Weight, lbs	2,200	2,400	2,200	2,400	2,900	3,100	2,900	3,100
Approximate Shipping Weight, lbs	2,350	2,550	2,350	2,550	3,100	3,300	3,100	3,300
Unit Dimensions, H in. x W in. x D in.	76.2 x 89.1 x 40.3	88.1 x 88.2 x 40.3 (91.1 x 88.2 x 40.3 w/ Hot Gas/Water Reheat)	76.0 x 87.3 x 38.4	88.1 x 88.2 x 40.3 (91.1 x 88.2 x 40.3 w/ Hot Gas/Water Reheat)	76.02 x 118.4 x 40.3	88.1 x 118.4 x 40.3 (91.1 x 118.4 x 40.3 w/ Hot Gas/ Water Reheat)	76.2 x 118.4 x 40.3	88.1 x 118.4 x 40.3 (91.1 x 118.4 x 40.3 w/ Hot Gas/ Water Reheat)
Shipping Dimensions, H in. x W in. x D in.	83 x 95 x 46	95 x 95 x 46 (98 x 95 x 46 w/ Hot Gas/Water Reheat)	83 x 95 x 46	95 x 95 x 46 (98 x 95 x 46 w/ Hot Gas/Water Reheat)	83 x 124 x 46	95 x 124 x 46 (98 x 124 x 46 w/ Hot Gas/Water Reheat)	83 x 124 x 46	95 x 124 x 46 (98 x 124 x 46 w/ Hot Gas/Water Reheat)

Performance / Capacity Data

Water Cooled Systems – Downflow / Upflow Models 021 – 042

Models:	CFD-021-D2W	CFU-021-D2W	CFD-028-D2W	CFU-028-D2W	CFD-035-D2W	CFU-035-D2W	CFD-042-D2W	CFU-042-D2W
Evaporator Blower / Motor - Backward Inclined Direct Driven EC								
Nominal Horsepower per Blower, hp	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.7
Rated Air Flow, CFM @ .3 in. H_2O ESP	6,000	6,000	6,000	6,000	6,000	6,000	8,000	8,000
Quantity of Blowers	1	1	1	1	1	1	2	2
Evaporator Coil - Aluminum Fin, Copper Tube								
Direct Expansion (DX) Coil								
Rows (Face Area, ft^2)	2 (21.6)	2 (21.6)	2 (21.6)	2 (21.6)	2 (21.6)	2 (21.6)	3 (24.7)	3 (24.7)
Face Velocity, fpm	278	278	278	278	278	278	323	323
Free Cooling (FC) Water Economizer / Alternate Water Source (AWS) Coil (Optional)								
Coil Rows (Face Area, ft^2)	4 (21.6)	4 (21.6)	4 (21.6)	4 (21.6)	4 (21.6)	4 (21.6)	4 (24.7)	4 (24.7)
Face Velocity, fpm	278	278	278	278	278	278	323	323
CV (FC Valve, 3-way, 400 PSI)	16	16	16	16	20	20	20	20
CV (AWS valve, 3-way, 400 PSI)	11	11	11	11	11	11	20	20
Compressors - R-410A								
Type (Qty)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)
Input per Compressor, W	2,970	2,970	3,630	3,630	4,250	4,250	4,660	4,660
Water Cooled Condensers - Micro-Plate (Based on 0% Glycol Solution)								
Total Heat of Rejection, kW (MBH)	39.1 (134)	39.1 (134)	45.1 (154)	45.1 (154)	49.9 (170)	49.9 (170)	60.1 (206)	60.1 (206)
Flow Rate, GPM @ 85°F EWT / 95°F LWT	27	27	31	31	35	35	42	42
Unit Pressure Drop, ft H_2O	15	15	18	18	21	21	15	15
Reheat - Performance Capacities DO NOT Include Evaporator Motor Heat @ Rated CFM & ESP								
Electric Reheat - Finned Tubular								
Number of Stages	1	1	1	1	1	1	2	2
Total Nominal Capacity, kW (MBH)	9.0 (30)	9.0 (30)	9.0 (30)	9.0 (30)	9.0 (30)	9.0 (30)	18.0 (61)	18.0 (61)
Hot Gas Reheat - with 3-Way Heat Reclaim Valve								
Total Capacity, kW (MBH)	2.9 (10)	2.9 (10)	3.6 (12)	3.6 (12)	4.2 (14)	4.2 (14)	4.6 (15)	4.6 (15)
Hot Water Reheat - Rated at 180°F EWT / 160°F LWT & 72°FDB/60.0°FWB EAT								
Total Capacity, kW (MBH)	19.4 (66)	23.8 (81)	19.4 (66)	23.8 (81)	19.4 (66)	23.8 (81)	24.2 (82)	29.8 (101)
Flow Rate, GPM (Pressure Drop, ft H_2O)	7 (4)	9 (5)	7 (4)	9 (5)	7 (4)	9 (5)	9 (5)	11 (8)
Control Valve	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating
Steam Reheat - Rated at 5 psig & 72°FDB EAT								
Total Capacity, kW (MBH)	44.5 (152)	61.9 (211)	44.5 (152)	61.9 (211)	44.5 (152)	61.9 (211)	58.2 (198)	79.2 (270)
Condensate, lbs/hr	158	220	158	220	158	220	207	281
Control Valve	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating
Humidification - Electrode Steam Canister with Adjustable Output								
Steam Output, lbs/hr (Power Input, kW)	4-15 (5.1)	4-15 (5.1)	4-15 (5.1)	4-15 (5.1)	4-15 (5.1)	4-15 (5.1)	4-15 (5.1)	4-15 (5.1)
Standard Control	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating
Filters - Disposable, MERV 8 Average Dust Spot Efficiency (Standard), MERV 11 Average Dust Spot Efficiency (Optional)								
Actual Size, H in. x W in. x D in. (Qty)	37.0 x 18.5 x 3.8 (3)	35.9 x 24.8 x 3.8 (1) 35.9 x 18.9 x 3.8 (1) 21.0 x 18.9 x 3.8 (1)	37.0 x 18.5 x 3.8 (3)	35.9 x 24.8 x 3.8 (1) 35.9 x 18.9 x 3.8 (1) 21.0 x 18.9 x 3.8 (1)	37.0 x 18.5 x 3.8 (3)	35.9 x 24.8 x 3.8 (1) 35.9 x 18.9 x 3.8 (1) 21.0 x 18.9 x 3.8 (1)	37.0 x 21.0 x 3.8 (3)	35.9 x 24.8 x 3.8 (2) 24.8 x 21.0 x 3.8 (1)
Connection Sizes - Copper								
Condenser In & Out, OD in.	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	2 1/8	2 1/8
AWS In & Out, OD in.	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	2 1/8	2 1/8
Condensate Drain without Pump, OD in.	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8
Condensate Drain with Pump, OD in.	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2
Humidifier Inlet, OD in.	1/4	1/4	1/4	1/4	1/4	1/4	1/4	1/4
Hot Water Reheat In & Out, OD in.	7/8	7/8	7/8	7/8	7/8	7/8	7/8	7/8
Physical Data								
Approximate Unit Weight, lbs	1,800	2,000	1,800	2,000	1,800	2,000	2,300	2,500
Approximate Shipping Weight, lbs	1,900	2,100	1,900	2,100	1,900	2,100	2,450	2,650
Unit Dimensions, H in. x W in. x D in.	76.2 x 77.4 x 40.3	88.1 x 77.4 x 40.3 (91.1 x 77.4 x 40.3 w/ Hot Gas/Water Reheat)	76.2 x 77.4 x 40.3	88.1 x 77.4 x 40.3 (91.1 x 77.4 x 40.3 w/ Hot Gas/Water Reheat)	76.2 x 77.4 x 40.3	88.1 x 77.4 x 40.3 (91.1 x 77.4 x 40.3 w/ Hot Gas/Water Reheat)	76.2 x 89.1 x 40.3	88.1 x 88.2 x 40.3 (91.1 x 88.2 x 40.3 w/ Hot Gas/Water Reheat
Shipping Dimensions, H in. x W in. x D in.	83.0 x 80.0 x 46.0	95.0 x 80.0 x 46.0 (98.0 x 80.0 x 46.0 w/ Hot Gas/Water Reheat)	83.0 x 80.0 x 46.0	95.0 x 80.0 x 46.0 (98.0 x 80.0 x 46.0 w/ Hot Gas/Water Reheat)	83.0 x 80.0 x 46.0	95.0 x 80.0 x 46.0 (98.0 x 80.0 x 46.0 w/ Hot Gas/Water Reheat)	83.0 x 95.0 x 46.0	95.0 x 95.0 x 46.0 (98.0 x 95.0 x 46.0 w/ Hot Gas/Water Reheat)

Performance / Capacity Data

Water Cooled Systems – Downflow / Upflow Models 053 – 105

Models	CFD-053-D2W	CFU-053-D2W	CFD-070-D2W	CFU-070-D2W	CFD-088-D2W	CFU-088-D2W	CFD-105-D2W	CFU-105-D2W
Evaporator Blower / Motor - Backward Inclined Direct Driven EC								
Nominal Horsepower per Blower, hp	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.7
Rated Air Flow, CFM @ .3 in. H2O ESP	9,000	9,000	10,000	10,000	15,750	15,750	15,750	15,750
Quantity of Blowers	2	2	2	2	3	3	3	3
Evaporator Coil - Aluminum Fin, Copper Tube								
Direct Expansion (DX) Coil								
Rows (Face Area, ft_2)	3 (24.7)	3 (24.7)	3 (24.7)	3 (24.7)	3 (39.8)	3 (39.8)	3 (39.8)	3 (39.8)
Face Velocity, fpm	364	364	404	404	395	395	395	395
Free Cooling (FC) Water Economizer / Alternate Water Source (AWS) Coil								
Coil Rows (Face Area, ft_2)	4 (24.7)	4 (24.7)	4 (24.7)	4 (24.7)	4 (39.8)	4 (39.8)	4 (39.8)	4 (39.8)
Face Velocity, fpm	364	364	404	404	395	395	395	395
CV (FC Valve, 3-way, 400 PSI)	28	28	35.4	35.4	48	48	48	48
CV (AWS valve, 3-way, 400 PSI)	20	20	20	20	28	28	28	28
Compressors - R-410A								
Type (Qty)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)
Input per Compressor, W	6,310	6,310	8,140	8,140	10,710	10,710	12,460	12,460
Water Cooled Condensers - Micro-Plate (Based on 0% Glycol Solution)								
Total Heat of Rejection, kW (MBH)	75.7 (259)	75.7 (259)	96.7 (330)	96.7 (330)	129.7 (443)	129.7 (443)	144.7 (494)	144.7 (494)
Flow Rate, GPM @ 85°F EWT / 95°F LWT	52	52	67	67	89	89	99	99
Unit Pressure Drop, ft H_2O	15	15	18	18	20	20	24	24
Reheat - Performance Capacities DO NOT Include Evaporator Motor Heat @ Rated CFM & ESP								
Electric Reheat - Finned Tubular								
Number of Stages	2	2	2	2	3	3	3	3
Total Nominal Capacity, kW (MBH)	18.0 (61)	18.0 (61)	18.0 (61)	18.0 (61)	27.0 (92)	27.0 (92)	27.0 (92)	27.0 (92)
Hot Gas Reheat - with 3-Way Heat Reclaim Valve								
Total Capacity, kW (MBH)	6.3 (21)	6.3 (21)	8.1 (27)	8.1 (27)	10.7 (36)	10.7 (36)	12.4 (42)	12.4 (42)
Hot Water Reheat - Rated at 180°F EWT / 160°F LWT & 72°FDB/60.0°FWB EAT								
Total Capacity, kW (MBH)	25.5 (87)	31.1 (106)	26.6 (91)	32.2 (110)	45 (153)	56.1 (191)	45 (153)	56.1 (191)
Flow Rate, GPM (Pressure Drop, ft H_2O)	9 (6)	11 (9)	10 (6)	12 (9)	16 (18)	20 (27)	16 (18)	20 (27)
Control Valve	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating
Steam Reheat - Rated at 5 psig & 72°FDB EAT								
Total Capacity, kW (MBH)	62.8 (214)	84.7 (289)	66.8 (228)	89.1 (304)	105.2 (359)	146.8 (501)	105.2 (359)	146.8 (501)
Condensate, lbs/hr	223	301	237	316	374	521	374	521
Control Valve	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating
Humidification - Electrode Steam Canister with Adjustable Output								
Steam Output, lbs/hr (Power Input, kW)	4-15 (5.1)	4-15 (5.1)	4-15 (5.1)	4-15 (5.1)	4-25 (8.5)	4-25 (8.5)	4-25 (8.5)	4-25 (8.5)
Standard Control	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating
Filters - Disposable, MERV 8 Average Dust Spot Efficiency (Standard), MERV 11 Average Dust Spot Efficiency (Optional)								
Actual Size, H in. x W in. x D in. (Qty)	37.0 x 21.0 x 3.8 (3)	35.9 x 24.8 x 3.8 (2) 24.8 x 21.0 x 3.8 (1)	37.0 x 21.0 x 3.8 (3)	35.9 x 24.8 x 3.8 (2) 24.8 x 21.0 x 3.8 (1)	37.0 x 18.5 x 3.8 (5)	35.9 x 24.8 x 3.8 (3) 24.8 x 21.0 x 3.8 (2)	37.0 x 18.5 x 3.8 (5)	35.9 x 24.8 x 3.8 (3) 24.8 x 21.0 x 3.8 (2)
Connection Sizes - Copper								
Condenser In & Out, OD in.	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8
AWS In & Out, OD in.	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8
Condensate Drain without Pump, OD in.	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8
Condensate Drain with Pump, OD in.	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2
Humidifier Inlet, OD in.	1/4	1/4	1/4	1/4	1/4	1/4	1/4	1/4
Hot Water Reheat In & Out, OD in.	7/8	7/8	7/8	7/8	7/8	7/8	7/8	7/8
Physical Data								
Approximate Unit Weight, lbs	2,300	2,500	2,300	2,500	3,000	3,200	3,000	3,200
Approximate Shipping Weight, lbs	2,450	2,650	2,450	2,650	3,200	3,400	3,200	3,400
Unit Dimensions, H in. x W in. x D in.	76.2 x 89.1 x 40.3	88.1 x 88.2 x 40.3 (91.1 x 88.2 x 40.3 w/ Hot Gas/Water Reheat)	76.2 x 89.1 x 40.3	88.1 x 88.2 x 40.3 (91.1 x 88.2 x 40.3 w/ Hot Gas/Water Reheat)	76.2 x 118.4 x 40.3	88.1 x 118.4 x 40.3 (91.1 x 118.4 x 40.3 w/ Hot Gas/ Water Reheat)	76.2 x 118.4 x 40.3	88.0 x 118.4 x 40.3 (91.0 x 118.4 x 40.3 w/ Hot Gas/ Water Reheat)
Shipping Dimensions, H in. x W in. x D in.	83.0 x 95.0 x 46.0	95.0 x 95.0 x 46.0 (98.0 x 95.0 x 46.0 w/ Hot Gas/Water Reheat)	83.0 x 95.0 x 46.0	95.0 x 95.0 x 46.0 (98.0 x 95.0 x 46.0 w/ Hot Gas/Water Reheat)	83.0 x 124.0 x 46.0	95.0 x 124.0 x 46.0 (98.0 x 124.0 x 46.0 w/ Hot Gas/Water Reheat)	83.0 x 124.0 x 46.0	95.0 x 124.0 x 46.0 (98.0 x 124.0 x 46.0 w/ Hot Gas/Water Reheat)

Performance / Capacity Data

Glycol Cooled Systems – Downflow / Upflow Models 021 – 042

Models:	CFD-021-D2G	CFU-021-D2G	CFD-028-D2G	CFU-028-D2G	CFD-035-D2G	CFU-035-D2G	CFD-042-D2G	CFU-042-D2G
Evaporator Blower / Motor - Backward Inclined Direct Driven EC								
Nominal Horsepower per Blower, hp	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.7
Rated Air Flow, CFM @ .3 in. H_2O ESP	6,000	6,000	6,000	6,000	6,000	6,000	8,000	8,000
Quantity of Blowers	1	1	1	1	1	1	2	2
Evaporator Coil - Aluminum Fin, Copper Tube								
Direct Expansion (DX) Coil								
Rows (Face Area, ft^2)	2 (21.6)	2 (21.6)	2 (21.6)	2 (21.6)	2 (21.6)	2 (21.6)	3 (24.7)	3 (24.7)
Face Velocity, fpm	278	278	278	278	278	278	323	323
Free Cooling (FC) Glycol Economizer / Alternate Water Source (AWS) Coil								
Coil Rows (Face Area, ft^2)	4 (21.6)	4 (21.6)	4 (21.6)	4 (21.6)	4 (21.6)	4 (21.6)	4 (24.7)	4 (24.7)
Face Velocity, fpm	278	278	278	278	278	278	323	323
CV (FC Valve, 3-way, 400 PSI)	16	16	16	16	20	20	20	20
CV (AWS valve, 3-way, 400 PSI)	11	11	11	11	11	11	20	20
Compressors - R-410A								
Type (Qty)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)	Scroll (2)
Input per Compressor, W	4,170	4,170	5,040	5,040	5,650	5,650	6,240	6,240
Glycol Cooled Condensers - Micro-Plate (Based on 40% Ethylene Glycol Solution)								
Total Heat of Rejection, kW (MBH)	37.9 (129)	37.9 (129)	43.8 (150)	43.8 (150)	47.9 (164)	47.9 (164)	57.6 (197)	57.6 (197)
Flow Rate, GPM @ 110°F EGT / 120°F LGT	29	29	34	34	36	36	44	44
Unit Pressure Drop, ft H_2O	17	17	21	21	24	24	16	16
Reheat / Heat - Performance Capacities DO NOT Include Evaporator Motor Heat @ Rated CFM & ESP								
Electric Reheat - Finned Tubular								
Number of Stages	1	1	1	1	1	1	2	2
Total Nominal Capacity, kW (MBH)	9.0 (30)	9.0 (30)	9.0 (30)	9.0 (30)	9.0 (30)	9.0 (30)	18.0 (61)	18.0 (61)
Hot Gas Reheat - with 3-Way Heat Reclaim Valve								
Total Capacity, kW (MBH)	4.1 (14)	4.1 (14)	5.0 (17)	5.0 (17)	5.6 (19)	5.6 (19)	6.2 (21)	6.2 (21)
Hot Water Reheat - Rated at 180°F EWT / 160°F LWT & 72°FDB/60.0°FWB EAT								
Total Capacity, kW (MBH)	19.4 (66)	23.8 (81)	19.4 (66)	23.8 (81)	19.4 (66)	23.8 (81)	24.2 (82)	29.8 (101)
Flow Rate, GPM (Pressure Drop, ft H_2O)	7 (4)	9 (5)	7 (4)	9 (5)	7 (4)	9 (5)	9 (5)	11 (8)
Control Valve	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating
Steam Reheat - Rated at 5 psig & 72°FDB EAT								
Total Capacity, kW (MBH)	44.5 (152)	61.9 (211)	44.5 (152)	61.9 (211)	44.5 (152)	61.9 (211)	58.2 (198)	79.2 (270)
Condensate, lbs/hr	158	220	158	220	158	220	207	281
Control Valve	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating
Humidification - Electrode Steam Canister with Adjustable Output								
Steam Output, lbs/hr / (Power Input, kW)	4-15 / (5.1)	4-15 / (5.1)	4-15 / (5.1)	4-15 / (5.1)	4-15 / (5.1)	4-15 / (5.1)	4-15 / (5.1)	4-15 / (5.1)
Control Type	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating
Filters - Disposable, MERV 8 Average Dust Spot Efficiency (Standard), MERV 11 Average Dust Spot Efficiency (Optional)								
Actual Size, H in. x W in. x D in. (Qty)	37.0 x 18.5 x 3.8 (3)	35.9 x 24.8 x 3.8 (1) 35.9 x 18.9 x 3.8 (1) 21.0 x 18.9 x 3.8 (1)	37.0 x 18.5 x 3.8 (3)	35.9 x 24.8 x 3.8 (1) 35.9 x 18.9 x 3.8 (1) 21.0 x 18.9 x 3.8 (1)	37.0 x 18.5 x 3.8 (3)	35.9 x 24.8 x 3.8 (1) 35.9 x 18.9 x 3.8 (1) 21.0 x 18.9 x 3.8 (1)	37.0 x 21.0 x 3.8 (3)	35.9 x 24.8 x 3.8 (2) 24.8 x 21.0 x 3.8 (1)
Connection Sizes - Copper								
Condenser In & Out, OD in.	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	2 1/8	2 1/8
AWS In & Out, OD in.	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	2 1/8	2 1/8
Condensate Drain without Pump, OD in.	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8
Condensate Drain with Pump, OD in.	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2
Humidifier Inlet, OD in.	1/4	1/4	1/4	1/4	1/4	1/4	1/4	1/4
Hot Water Reheat In & Out, OD in.	7/8	7/8	7/8	7/8	7/8	7/8	7/8	7/8
Physical Data								
Approximate Unit Weight, lbs	1,800	2,000	1,800	2,000	1,800	2,000	2,300	2,500
Approximate Shipping Weight, lbs	1,900	2,100	1,900	2,100	1,900	2,100	2,450	2,650
Unit Dimensions, H in. x W in. x D in.	76.2 x 77.4 x 40.3	88.1 x 77.4 x 40.3 (91.1 x 77.4 x 40.3 w/ Hot Gas/Water Reheat)	76.2 x 77.4 x 40.3	88.1 x 77.4 x 40.3 (91.1 x 77.4 x 40.3 w/ Hot Gas/Water Reheat)	76.2 x 77.4 x 40.3	88.1 x 77.4 x 40.3 (91.1 x 77.4 x 40.3 w/ Hot Gas/Water Reheat)	76.0 x 89.1 x 38.4	88.1 x 89.1 x 40.3 (91.1 x 89.1 x 40.3 w/ Hot Gas/Water Reheat)
Shipping Dimensions, H in. x W in. x D in.	83.0 x 80.0 x 46.0	95.0 x 80.0 x 46.0 (98.0 x 80.0 x 46.0 w/ Hot Gas/Water Reheat)	83.0 x 80.0 x 46.0	95.0 x 80.0 x 46.0 (98.0 x 80.0 x 46.0 w/ Hot Gas/Water Reheat)	83.0 x 80.0 x 46.0	95.0 x 80.0 x 46.0 (98.0 x 80.0 x 46.0 w/ Hot Gas/Water Reheat)	83.0 x 95.0 x 46.0	95.0 x 95.0 x 46.0 (98.0 x 95.0 x 46.0 w/ Hot Gas/Water Reheat)

Performance / Capacity Data

Glycol Cooled Systems – Downflow / Upflow Models 053 – 105

Models:	CFD-053-D2G	CFU-053-D2G	CFD-070-D2G	CFU-070-D2G	CFD-088-D2G	CFU-088-D2G	CFD-105-D2G	CFU-105-D2G
Evaporator Blower / Motor - Backward Inclined Direct Driven EC								
Nominal Horsepower per Blower, hp	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.7
Rated Air Flow, CFM @ .3 in. H_2O ESP	9,000	9,000	10,000	10,000	15,750	15,750	15,750	15,750
Quantity of Blowers	2	2	2	2	3	3	3	3
Evaporator Coil - Aluminum Fin, Copper Tube								
Direct Expansion (DX) Coil								
Rows (Face Area, ft^2)	3 (24.7)	3 (24.7)	3 (24.7)	3 (24.7)	3 (39.8)	3 (39.8)	3 (39.8)	3 (39.8)
Face Velocity, fpm	364	364	404	404	395	395	395	395
Free Cooling (FC) Glycol Economizer / Alternate Water Source (AWS) Coil								
Coil Rows (Face Area, ft^2)	4 (24.7)	4 (24.7)	4 (24.7)	4 (24.7)	4 (39.8)	4 (39.8)	4 (39.8)	4 (39.8)
Face Velocity, fpm	364	364	404	404	395	395	395	395
CV (FC Valve, 3-way, 400 PSI)	28	28	35.4	35.4	48	48	48	48
CV (AWS valve, 3-way, 400 PSI)	20	20	20	20	28	28	28	28
Compressors - R-410A								
Type / (Qty)	Scroll / (2)	Scroll / (2)	Scroll / (2)	Scroll / (2)	Scroll / (2)	Scroll / (2)	Scroll / (2)	Scroll / (2)
Input per Compressor, W	8,370	8,370	11,070	11,070	14,090	14,090	16,600	16,600
Glycol Cooled Condensers - Micro-Plate (Based on 40% Ethylene Glycol Solution)								
Total Heat of Rejection, kW (MBH)	73.2 (250)	73.2 (250)	92.4 (316)	92.4 (316)	124.7 (426)	124.7 (426)	138.8 (474)	138.8 (474)
Flow Rate, GPM @ 110°F EGT / 120°F LGT	56	56	70	70	95	95	105	105
Unit Pressure Drop, ft H_2O	17	17	20	20	23	23	28	28
Reheat - Performance Capacities DO NOT Include Evaporator Motor Heat @ Rated CFM & ESP								
Electric Reheat - Finned Tubular								
Number of Stages	2	2	2	2	3	3	3	3
Total Nominal Capacity, kW (MBH)	18.0 (61)	18.0 (61)	18.0 (61)	18.0 (61)	27.0 (92)	27.0 (92)	27.0 (92)	27.0 (92)
Hot Gas Reheat - with 3-Way Heat Reclaim Valve								
Total Capacity, kW (MBH)	8.3 (28)	8.3 (28)	11.0 (37)	11.0 (37)	14.0 (48)	14.0 (48)	16.6 (56)	16.6 (56)
Hot Water Reheat - Rated at 180°F EWT / 160°F LWT & 72°FDB/60.0°FWB EAT								
Total Capacity, kW (MBH)	25.5 (87)	31.1 (106)	26.6 (91)	32.2 (110)	45 (153)	56.1 (191)	45 (153)	56.1 (191)
Flow Rate, GPM (Pressure Drop, ft H_2O)	9 (6)	11 (9)	10 (6)	12 (9)	16 (18)	20 (27)	16 (18)	20 (27)
Control Valve	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating
Steam Reheat - Rated at 5 psig & 72°FDB EAT								
Total Capacity, kW (MBH)	62.8 (214)	84.7 (289)	66.8 (228)	89.1 (304)	105.2 (359)	146.8 (501)	105.2 (359)	146.8 (501)
Condensate, lbs/hr	223	301	237	316	374	521	374	521
Control Valve	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating
Humidification - Electrode Steam Canister with Adjustable Output								
Steam Output, lbs/hr (Power Input, kW)	4-15 (5.1)	4-15 (5.1)	4-15 (5.1)	4-15 (5.1)	4-25 (8.5)	4-25 (8.5)	4-25 (8.5)	4-25 (8.5)
Control Type	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating	Modulating
Filters - Disposable, MERV 8 Average Dust Spot Efficiency (Standard), MERV 11 Average Dust Spot Efficiency (Optional)								
Actual Size, H in. x W in. x D in. (Qty)	37.0 x 21.0 x 3.8 (3)	35.9 x 24.8 x 3.8 (2) 24.8 x 21.0 x 3.8 (1)	37.0 x 21.0 x 3.8 (3)	35.9 x 24.8 x 3.8 (2) 24.8 x 21.0 x 3.8 (1)	37.0 x 18.5 x 3.8 (5)	35.9 x 24.8 x 3.8 (3) 24.8 x 21.0 x 3.8 (2)	37.0 x 18.5 x 3.8 (5)	35.9 x 24.8 x 3.8 (3) 24.8 x 21.0 x 3.8 (2)
Connection Sizes - Copper								
Condenser In & Out, OD in.	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8
AWS In & Out, OD in.	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8	2 1/8
Condensate Drain without Pump, OD in.	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8	1 1/8
Condensate Drain with Pump, OD in.	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2
Humidifier Inlet, OD in.	1/4	1/4	1/4	1/4	1/4	1/4	1/4	1/4
Hot Water Reheat In & Out, OD in.	7/8	7/8	7/8	7/8	7/8	7/8	7/8	7/8
Physical Data								
Approximate Unit Weight, lbs	2,300	2,500	2,300	2,500	3,000	3,200	3,000	3,200
Approximate Shipping Weight, lbs	2,450	2,650	2,450	2,650	3,200	3,400	3,200	3,400
Unit Dimensions, H in. x W in. x D in.	76.2 x 89.1 x 40.3	88.1 x 89.1 x 40.3 (91.1 x 89.1 x 40.3 w/ Hot Gas / Water Reheat)	76.2 x 89.1 x 40.3	88.1 x 89.1 x 40.3 (91.1 x 89.1 x 40.3 w/ Hot Gas / Water Reheat)	76.2 x 118.4 x 40.3	88.1 x 118.4 x 40.3 (91.1 x 118.4 x 40.3 w/ Hot Gas / Water Reheat)	76.2 x 118.4 x 40.3	88.1 x 118.4 x 40.3 (91.1 x 118.4 x 40.3 w/ Hot Gas/Water Reheat)
Shipping Dimensions, H in. x W in. x D in.	83.0 x 95.0 x 46.0	95.0 x 95.0 x 46.0 (98.0 x 95.0 x 46.0 w/ Hot Gas/Water Reheat)	83.0 x 95.0 x 46.0	95.0 x 95.0 x 46.0 (98.0 x 95.0 x 46.0 w/ Hot Gas/Water Reheat)	83.0 x 124.0 x 46.0	95.0 x 124.0 x 46.0 (98.0 x 124.0 x 46.0 w/ Hot Gas / Water Reheat)	83.0 x 124.0 x 46.0	95.0 x 124.0 x 46.0 (98.0 x 124.0 x 46.0 w/ Hot Gas / Water Reheat)

Performance / Capacity Data

Air Cooled Systems – Downflow / Upflow Models 021 – 105

Models: CFD/U-	021-D2A	028-D2A	035-D2A	042-D2A	053-D2A	070-D2A	088-D2A	105-D2A
NET DX COOLING CAPACITY - (Includes Motor Heat @ Rated CFM & ESP)								
80°FDB/62.9°FWB, 38% RH, 52.3°FDP								
Total, kW (MBH)	27.1 (92)	31.4 (107)	34.4 (117)	40.1 (136)	51.6 (176)	65.6 (224)	89.7 (306)	99.1 (338)
Sensible, kW (MBH)	27.1 (92)	31.4 (107)	34.4 (117)	40.1 (136)	51.6 (176)	64.1 (218)	89.7 (306)	99.1 (338)
72°FDB/60.0°FWB, 50% RH, 52.3°FDP								
Total, kW (MBH)	24.0 (82)	28.5 (97)	31.8 (108)	35.1 (120)	46.4 (158)	62.1 (212)	80.9 (276)	92.4 (315)
Sensible, kW (MBH)	23.2 (79)	27.8 (95)	30.2 (103)	33.6 (114)	42.0 (143)	58.3 (199)	77.3 (263)	88.1 (300)
NET AWS COOLING CAPACITY - Based on 45°F EWT / 55°F LWT, 0% Glycol Solution (Includes Motor Heat @ Rated CFM & ESP)								
80°FDB/62.9°FWB, 38% RH, 52.3°FDP								
Total, kW (MBH)	41.3 (141)	41.3 (141)	41.3 (141)	51.6 (176)	56.3 (192)	60.6 (207)	97.8 (333)	97.8 (333)
Sensible, kW (MBH)	41.3 (141)	41.3 (141)	41.3 (141)	51.6 (176)	56.3 (192)	60.6 (207)	97.8 (333)	97.8 (333)
Flow Rate, GPM (Unit Press. Drop, ft H2O)	31 (28)	31 (28)	31 (28)	40 (21)	43 (23)	46 (26)	74 (36)	74 (36)
72°FDB/60.0°FWB, 50% RH, 52.3°FDP								
Total, kW (MBH)	27.2 (93)	27.2 (93)	27.2 (93)	33.7 (115)	36.9 (125)	39.8 (136)	64.2 (219)	64.2 (219)
Sensible, kW (MBH)	27.2 (93)	27.2 (93)	27.2 (93)	33.7 (115)	36.9 (125)	39.8 (136)	64.2 (219)	64.2 (219)
Flow Rate, GPM / (Unit Press. Drop, ft H2O)	21 / (15)	21 / (15)	21 / (15)	28 / (13)	30 / (14)	32 / (15)	51 / (20)	51 / (20)

Performance / Capacity Data

Water Cooled Systems – Downflow / Upflow Models 021 – 105

Models: CFD/U-	021-D2W	028-D2W	035-D2W	042-D2W	053-D2W	070-D2W	088-D2W	105-D2W
NET DX COOLING CAPACITY - (Includes Motor Heat @ Rated CFM & ESP) 80°FDB/62.9°FWB, 38% RH, 52.3°FDP								
Total, kW (MBH)	29.9 (102)	34.7 (118)	38.1 (130)	44.5 (152)	56.8 (194)	74.1 (252)	98.9 (337)	110.4 (376)
Sensible, kW (MBH)	29.9 (102)	34.7 (118)	38.1 (130)	44.5 (152)	56.8 (194)	73.5 (250)	98.9 (337)	110.4 (376)
72°FDB/60.0°FWB, 50% RH, 52.3°FDP								
Total, kW (MBH)	26.9 (91)	32.3 (110)	35.8 (122)	39.9 (136)	52.9 (180)	69.7 (237)	92.2 (314)	104.8 (357)
Sensible, kW (MBH)	26.9 (91)	30.5 (104)	30.4 (103)	39.0 (133)	50.2 (171)	54.1 (184)	87.9 (299)	93.1 (317)
NET FC COOLING CAPACITY - Based on 45°F EWT, 0% Glycol Solution (Includes Motor Heat @ Rated CFM & ESP)								
80°FDB/62.9°FWB, 38% RH, 52.3°FDP								
Total, kW (MBH)	39.9 (136)	41.6 (142)	43.0 (146)	52.5 (179)	59.3 (202)	66.1 (225)	102.9 (351)	104.9 (357)
Sensible, kW (MBH)	39.9 (136)	41.6 (142)	43.0 (146)	52.5 (179)	59.3 (202)	66.1 (225)	102.9 (351)	104.9 (357)
Flow Rate, GPM (Unit Press. Drop, ft H_2O)	27 (26)	31 (33)	35 (36)	42 (34)	52 (35)	67 (46)	89 (46)	99 (56)
72°FDB/60.0°FWB, 50% RH, 52.3°FDP								
Total, kW (MBH)	30.2 (103)	34.1 (116)	35.0 (119)	42.5 (145)	47.6 (162)	52.8 (180)	83.0 (283)	84.2 (287)
Sensible, kW (MBH)	30.2 (103)	32.5 (111)	33.2 (113)	40.6 (138)	46.0 (157)	51.4 (175)	80.7 (275)	81.7 (278)
Flow Rate, GPM (Unit Press. Drop, ft H_2O)	27 (26)	31 (33)	35 (36)	42 (34)	52 (35)	67 (46)	89 (46)	99 (56)
NET AWS COOLING CAPACITY - Based on 45°F EWT / 55°F LWT, 0% Glycol Solution (Includes Motor Heat @ Rated CFM & ESP)								
80°FDB/62.9°FWB, 38% RH, 52.3°FDP								
Total, kW (MBH)	41.3 (141)	41.3 (141)	41.3 (141)	51.6 (176)	56.3 (192)	60.6 (207)	97.8 (333)	97.8 (333)
Sensible, kW (MBH)	41.3 (141)	41.3 (141)	41.3 (141)	51.6 (176)	56.3 (192)	60.6 (207)	97.8 (333)	97.8 (333)
Flow Rate, GPM (Unit Press. Drop, ft H_2O)	31 (28)	31 (28)	31 (28)	40 (21)	43 (23)	46 (26)	74 (36)	74 (36)
72°FDB/60.0°FWB, 50% RH, 52.3°FDP								
Total, kW (MBH)	27.2 (93)	27.2 (93)	27.2 (93)	33.7 (115)	36.9 (125)	39.8 (136)	64.2 (219)	64.2 (219)
Sensible, kW (MBH)	27.2 (93)	27.2 (93)	27.2 (93)	33.7 (115)	36.9 (125)	39.8 (136)	64.2 (219)	64.2 (219)
Flow Rate, GPM (Unit Press. Drop, ft H_2O)	21 (15)	21 (15)	21 (15)	28 (13)	30 (14)	32 (15)	51 (20)	51 (20)

Performance / Capacity Data

Glycol Cooled Systems – Downflow / Upflow Models 021 – 105

Models: CFD/U-	021-D2G	028-D2G	035-D2G	042-D2G	053-D2G	070-D2G	088-D2G	105-D2G
NET DX COOLING CAPACITY - (Includes Motor Heat @ Rated CFM & ESP)								
80°FDB/62.9°FWB, 38% RH, 52.3°FDP								
Total, kW (MBH)	26.3 (89)	30.5 (104)	33.4 (114)	38.8 (132)	50.1 (171)	63.9 (218)	87.1 (297)	96.2 (328)
Sensible, kW (MBH)	26.3 (89)	30.5 (104)	33.4 (114)	38.8 (132)	50.1 (171)	63.9 (218)	87.1 (297)	96.2 (328)
72°FDB/60.0°FWB, 50% RH, 52.3°FDP								
Total, kW (MBH)	23.2 (79)	27.4 (93)	30.7 (104)	34.0 (116)	44.9 (153)	59.8 (204)	78.3 (267)	89.0 (303)
Sensible, kW (MBH)	23.2 (79)	27.2 (92)	29.5 (100)	34.0 (116)	44.3 (151)	57.0 (194)	78.3 (267)	85.9 (293)
NET FC COOLING CAPACITY - Based on 45°F EGT, 40% Ethylene Glycol Solution (Includes Motor Heat @ Rated CFM & ESP)								
80°FDB/62.9°FWB, 38% RH, 52.3°FDP								
Total, kW (MBH)	30.8 (105)	32.5 (111)	33.1 (113)	41.9 (143)	49.4 (168)	56.7 (193)	84.3 (287)	88.0 (300)
Sensible, kW (MBH)	30.8 (105)	32.5 (111)	33.1 (113)	41.9 (143)	49.4 (168)	56.7 (193)	84.3 (287)	88.0 (300)
Flow Rate, GPM (Unit Press. Drop, ft H_2O)	29 (31)	34 (39)	36 (40)	44 (38)	56 (42)	70 (56)	95 (56)	105 (69)
72°FDB/60.0°FWB, 50% RH, 52.3°FDP								
Total, kW (MBH)	23.1 (78)	24.4 (83)	24.9 (85)	30.9 (105)	36.7 (125)	42.4 (144)	62.9 (214)	65.8 (224)
Sensible, kW (MBH)	23.1 (78)	24.4 (83)	24.9 (85)	30.9 (105)	36.7 (125)	42.4 (144)	62.9 (214)	65.8 (224)
Flow Rate, GPM (Unit Press. Drop, ft H_2O)	29 (31)	34 (39)	36 (40)	44 (38)	56 (42)	70 (56)	95 (56)	105 (68)
NET AWS COOLING CAPACITY - Based on 45°F EWT / 55°F LWT, 0% Glycol Solution (Includes Motor Heat @ Rated CFM & ESP)								
80°FDB/62.9°FWB, 38% RH, 52.3°FDP								
Total, kW (MBH)	41.3 (141)	41.3 (141)	41.3 (141)	51.6 (176)	56.3 (192)	60.6 (207)	97.8 (333)	97.8 (333)
Sensible, kW (MBH)	41.3 (141)	41.3 (141)	41.3 (141)	51.6 (176)	56.3 (192)	60.6 (207)	97.8 (333)	97.8 (333)
Flow Rate, GPM (Unit Press. Drop, ft H_2O)	31 (28)	31 (28)	31 (28)	40 (21)	43 (23)	46 (26)	74 (36)	74 (36)
72°FDB/60.0°FWB, 50% RH, 52.3°FDP								
Total, kW (MBH)	27.2 (93)	27.2 (93)	27.2 (93)	33.7 (115)	36.9 (125)	39.8 (136)	64.2 (219)	64.2 (219)
Sensible, kW (MBH)	27.2 (93)	27.2 (93)	27.2 (93)	33.7 (115)	36.9 (125)	39.8 (136)	64.2 (219)	64.2 (219)
Flow Rate, GPM (Unit Press. Drop, ft H_2O)	21 (15)	21 (15)	21 (15)	28 (13)	30 (14)	32 (15)	51 (20)	51 (20)

Electrical Data — Dual Compressor Systems (Air, Water, Glycol)

Voltage	Condensate Pump	CFU/D-021		CFU/D-028		CFU/D-035		CFU/D-042		CFU/D-053		CFU/D-070		CFU/D-088		CFU/D-105	
		MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
Cooling w/ Electric Reheat & Humidification																	
208/3/60	No	58.1	70.0	60.6	70.0	69.6	80.0	100.4	110.0	106.9	125.0	114.0	125.0	-	-	-	-
208/3/60	Yes	60.2	70.0	62.7	70.0	71.7	90.0	102.5	110.0	109.0	125.0	116.1	125.0	-	-	-	-
460/3/60	No	26.7	30.0	28.9	35.0	29.8	35.0	50.9	60.0	53.6	60.0	60.0	70.0	85.3	100.0	86.1	100.0
460/3/60	Yes	27.5	30.0	29.7	35.0	30.6	35.0	51.7	60.0	54.4	60.0	60.8	70.0	86.1	100.0	86.9	100.0
575/3/60	No	22.6	25.0	22.3	25.0	23.9	30.0	39.6	40.0	42.0	45.0	46.0	50.0	69.9	80.0	74.6	90.0
575/3/60	Yes	23.2	25.0	22.9	25.0	24.6	30.0	40.3	45.0	42.7	45.0	46.7	50.0	70.5	80.0	75.3	90.0
Cooling w/ Electric Reheat Only (No Humidifier)																	
208/3/60	No	54.9	60.0	56.3	60.0	61.3	70.0	100.4	110.0	106.9	110.0	114.0	125.0	-	-	-	-
208/3/60	Yes	57.0	60.0	58.4	60.0	63.4	70.0	102.5	110.0	109.0	125.0	116.1	125.0	-	-	-	-
460/3/60	No	26.7	30.0	28.6	30.0	29.1	30.0	50.9	60.0	53.6	60.0	60.0	70.0	85.3	90.0	86.1	100.0
460/3/60	Yes	27.5	30.0	29.4	30.0	29.9	35.0	51.7	60.0	54.4	60.0	60.8	70.0	86.1	100.0	86.9	100.0
575/3/60	No	22.5	25.0	22.3	25.0	23.3	25.0	39.6	40.0	42.0	45.0	46.0	50.0	69.9	80.0	74.6	90.0
575/3/60	Yes	23.1	25.0	22.9	25.0	23.9	25.0	40.3	45.0	42.7	45.0	46.7	50.0	70.5	80.0	75.3	90.0
Cooling w/ Humidification (No Electric Reheat)																	
208/3/60	No	58.1	70.0	60.6	70.0	69.6	80.0	86.0	100.0	97.7	125.0	110.5	125.0	-	-	-	-
208/3/60	Yes	60.2	70.0	62.7	70.0	71.7	90.0	88.1	100.0	99.8	125.0	112.6	125.0	-	-	-	-
460/3/60	No	25.5	30.0	28.9	35.0	29.8	35.0	39.9	50.0	44.9	50.0	56.3	70.0	77.2	100.0	78.6	100.0
460/3/60	Yes	26.3	30.0	29.7	35.0	30.6	35.0	40.7	50.0	45.7	50.0	57.1	70.0	78.0	100.0	79.4	100.0
575/3/60	No	22.6	25.0	22.1	25.0	23.9	30.0	30.2	35.0	34.4	40.0	41.6	50.0	64.9	80.0	73.5	90.0
575/3/60	Yes	23.2	25.0	22.8	25.0	24.6	30.0	30.8	35.0	35.1	40.0	42.3	50.0	65.6	80.0	74.1	90.0
Cooling Only																	
208/3/60	No	43.9	50.0	46.4	60.0	55.4	70.0	71.8	90.0	82.5	110.0	96.3	125.0	146.9	175.0	157.0	200.0
208/3/60	Yes	46.0	60.0	48.5	60.0	57.5	70.0	73.9	90.0	85.6	110.0	98.4	125.0	149.0	200.0	159.1	200.0
460/3/60	No	18.9	25.0	22.2	30.0	23.1	30.0	33.2	40.0	38.2	50.0	49.7	60.0	66.1	80.0	67.4	90.0
460/3/60	Yes	19.7	25.0	23.0	30.0	23.9	30.0	34.0	40.0	39.0	50.0	50.5	60.0	66.9	80.0	68.2	90.0
575/3/60	No	17.3	20.0	16.8	20.0	18.6	25.0	24.8	30.0	29.1	35.0	36.3	45.0	56.0	70.0	64.6	80.0
575/3/60	Yes	17.9	20.0	17.4	20.0	19.2	25.0	25.5	30.0	29.8	35.0	37.0	45.0	56.7	70.0	65.2	80.0

NOTE: High short circuit current rating available on most models. For voltages not shown, call STULZ.

Electrical Data — Tandem Compressor Systems (Air, Water, Glycol)

Voltage	Condensate Pump	CFU/D-053		CFU/D-070		CFU/D-088		CFU/D-105	
		MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
Cooling w/ Electric Reheat & Humidification									
208/3/60	No	106.9	125.0	116.4	125.0	-	-	-	-
208/3/60	Yes	109.0	125.0	118.5	150.0	-	-	-	-
460/3/60	No	53.6	60.0	60.0	70.0	85.3	100.0	86.1	100.0
460/3/60	Yes	54.4	60.0	60.8	70.0	86.1	100.0	86.9	100.0
575/3/60	No	42.0	45.0	46.0	50.0	69.9	80.0	74.6	90.0
575/3/60	Yes	42.7	45.0	46.7	50.0	70.5	80.0	75.3	90.0
Cooling w/ Electric Reheat Only (No Humidifier)									
208/3/60	No	106.9	110.0	114.0	125.0	-	-	-	-
208/3/60	Yes	109.0	125.0	116.1	125.0	-	-	-	-
460/3/60	No	53.6	60.0	60.0	70.0	85.3	90.0	86.1	100.0
460/3/60	Yes	54.4	60.0	60.8	70.0	86.1	100.0	86.9	100.0
575/3/60	No	42.0	45.0	46.0	50.0	69.9	80.0	74.6	90.0
575/3/60	Yes	42.7	45.0	46.7	50.0	70.5	80.0	75.3	90.0
Cooling w/ Humidification (No Electric Reheat)									
208/3/60	No	99.1	125.0	116.4	125.0	-	-	-	-
208/3/60	Yes	101.2	125.0	118.5	150.0	-	-	-	-
460/3/60	No	44.7	50.0	54.8	70.0	79.7	100.0	80.5	100.0
460/3/60	Yes	45.5	50.0	55.6	70.0	80.5	100.0	81.3	100.0
575/3/60	No	36.8	45.0	42.0	50.0	64.2	80.0	69.0	90.0
575/3/60	Yes	37.5	45.0	42.7	50.0	64.9	80.0	69.6	90.0
Cooling Only									
208/3/60	No	84.9	110.0	102.5	125.0	145.6	175.0	156.4	200.0
208/3/60	Yes	87.0	110.0	104.3	125.0	147.7	175.0	158.5	200.0
460/3/60	No	38.0	50.0	48.2	60.0	68.6	90.0	69.3	90.0
460/3/60	Yes	38.8	50.0	49.0	60.0	69.4	90.0	70.1	90.0
575/3/60	No	31.5	40.0	36.7	45.0	55.3	70.0	60.1	80.0
575/3/60	Yes	32.2	40.0	37.4	50.0	56.0	70.0	60.7	80.0

NOTE: High short circuit current rating available on most models. For voltages not shown, contact STULZ.

Electrical Data — Advanced Humidification

Dual Compressor Systems (Air, Water, Glycol)

Voltage	Condensate Pump	CFD-021		CFD-028		CFD-035		CFD-042		CFD-053		CFD-070		CFD-088		CFD-105	
		MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
Cooling w/ Electric Reheat & Humidification																	
208/3/60	No	69.4	70.0	71.9	80.0	80.9	90.0	122.8	125.0	134.5	150.0	147.3	150.0	-	-	-	-
208/3/60	Yes	71.5	80.0	74.0	80.0	83.0	90.0	124.9	125.0	136.6	150.0	149.4	150.0	-	-	-	-
460/3/60	No	33.0	35.0	36.4	40.0	37.3	40.0	61.5	70.0	66.4	70.0	77.9	90.0	108.4	110.0	109.8	125.0
460/3/60	Yes	33.8	35.0	37.2	40.0	38.1	40.0	62.3	70.0	67.2	70.0	78.7	90.0	109.2	110.0	110.6	125.0
575/3/60	No	28.5	30.0	28.1	30.0	29.9	30.0	47.3	50.0	51.6	60.0	58.8	60.0	89.8	100.0	98.3	110.0
575/3/60	Yes	29.1	30.0	28.7	30.0	30.5	35.0	48.0	50.0	52.3	60.0	59.5	60.0	90.4	100.0	99.0	110.0
Cooling w/ Electric Reheat Only (No Humidifier)																	
208/3/60	No	69.4	70.0	71.9	80.0	80.9	90.0	122.8	125.0	134.5	150.0	147.3	150.0	-	-	-	-
208/3/60	Yes	71.5	80.0	74.0	80.0	83.0	90.0	124.9	125.0	136.6	150.0	149.4	150.0	-	-	-	-
460/3/60	No	33.0	35.0	36.4	40.0	37.3	40.0	61.5	70.0	66.4	70.0	77.9	90.0	108.4	110.0	109.8	125.0
460/3/60	Yes	33.8	35.0	37.2	40.0	38.1	40.0	62.3	70.0	67.2	70.0	78.7	90.0	109.2	110.0	110.6	125.0
575/3/60	No	28.5	30.0	28.1	30.0	29.9	30.0	47.3	50.0	51.6	60.0	58.8	60.0	89.8	100.0	98.3	110.0
575/3/60	Yes	29.1	30.0	28.7	30.0	30.5	35.0	48.0	50.0	52.3	60.0	59.5	60.0	90.4	100.0	99.0	110.0
Cooling w/ Humidification (No Electric Reheat)																	
208/3/60	No	58.1	70.0	60.6	70.0	69.6	80.0	86.0	100.0	97.7	125.0	110.5	125.0	-	-	-	-
208/3/60	Yes	60.2	70.0	62.7	70.0	71.7	90.0	88.1	110.0	99.8	125.0	112.6	125.0	-	-	-	-
460/3/60	No	25.5	30.0	28.9	35.0	29.8	35.0	39.9	50.0	44.9	50.0	56.3	70.0	77.2	100.0	78.6	100.0
460/3/60	Yes	26.3	30.0	29.7	35.0	30.6	35.0	40.7	50.0	45.7	50.0	57.1	70.0	78.0	100.0	79.4	100.0
575/3/60	No	22.6	25.0	22.1	25.0	23.9	30.0	30.2	35.0	34.4	40.0	41.6	50.0	64.9	80.0	73.5	90.0
575/3/60	Yes	23.2	25.0	22.8	25.0	24.6	30.0	30.8	35.0	35.1	40.0	42.3	50.0	65.6	80.0	74.1	90.0
Cooling Only																	
208/3/60	No	43.9	50.0	46.4	60.0	55.4	70.0	71.8	90.0	83.5	110.0	96.3	125.0	146.9	175.0	157.0	200.0
208/3/60	Yes	46.0	60.0	48.5	60.0	57.5	70.0	73.9	90.0	85.6	110.0	98.4	125.0	149.0	200.0	159.1	200.0
460/3/60	No	18.9	25.0	22.2	30.0	23.1	30.0	33.2	40.0	38.2	50.0	49.7	60.0	66.1	80.0	67.4	90.0
460/3/60	Yes	19.7	25.0	23.0	30.0	23.9	30.0	34.0	40.0	39.0	50.0	50.5	60.0	66.9	80.0	68.2	90.0
575/3/60	No	17.3	20.0	16.8	20.0	18.6	25.0	24.8	30.0	29.1	35.0	36.3	45.0	56.0	70.0	64.6	80.0
575/3/60	Yes	17.9	20.0	17.4	20.0	19.2	25.0	25.5	30.0	29.8	35.0	37.0	45.0	56.7	70.0	65.2	80.0

NOTE: High short circuit current rating available on most models. For voltages not shown, call STULZ.

Electrical Data — Advanced Humidification

Tandem Compressor Systems (Air, Water, Glycol)

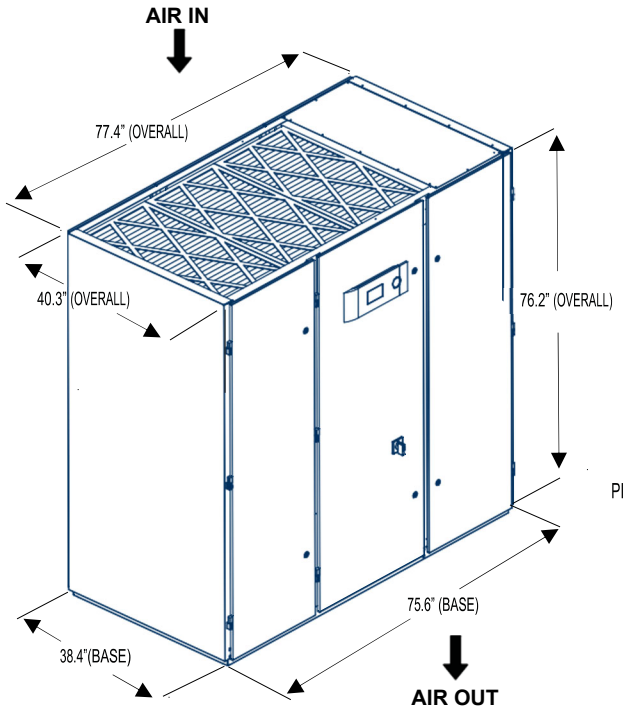
Voltage	Condensate Pump	CFU/D-053		CFU/D-070		CFU/D-088		CFU/D-105	
		MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
Cooling w/ Electric Reheat & Humidification									
208/3/60	No	135.9	150.0	153.2	175.0	-	-	-	-
208/3/60	Yes	138.0	150.0	155.3	175.0	-	-	-	-
460/3/60	No	66.2	70.0	76.4	80.0	110.9	125.0	111.7	125.0
460/3/60	Yes	67.0	70.0	77.2	80.0	111.7	125.0	112.5	125.0
575/3/60	No	54.0	60.0	59.2	60.0	89.1	100.0	93.8	110.0
575/3/60	Yes	54.7	60.0	59.9	60.0	89.7	100.0	94.5	110.0
Cooling w/ Electric Reheat Only (No Humidifier)									
208/3/60	No	135.9	150.0	153.2	175.0	-	-	-	-
208/3/60	Yes	138.0	150.0	155.3	175.0	-	-	-	-
460/3/60	No	66.2	70.0	76.4	80.0	110.9	125.0	111.7	125.0
460/3/60	Yes	67.0	70.0	77.2	80.0	111.7	125.0	112.5	125.0
575/3/60	No	54.0	60.0	59.2	60.0	89.1	100.0	93.8	110.0
575/3/60	Yes	54.7	60.0	59.9	60.0	89.7	100.0	94.5	110.0
Cooling w/ Humidification (No Electric Reheat)									
208/3/60	No	99.1	125.0	116.4	125.0	-	-	-	-
208/3/60	Yes	101.2	125.0	118.5	150.0	-	-	-	-
460/3/60	No	44.7	50.0	54.8	70.0	79.7	100.0	80.5	100.0
460/3/60	Yes	45.5	50.0	55.6	70.0	80.5	100.0	81.3	100.0
575/3/60	No	36.8	45.0	42.0	50.0	64.2	80.0	69.0	90.0
575/3/60	Yes	37.5	45.0	42.7	50.0	64.9	80.0	69.6	90.0
Cooling Only									
208/3/60	No	84.9	110.0	102.2	125.0	145.6	175.0	156.4	200.0
208/3/60	Yes	87.0	110.0	104.3	125.0	147.7	175.0	158.5	200.0
460/3/60	No	38.0	50.0	48.2	60.0	68.6	90.0	69.3	90.0
460/3/60	Yes	38.8	50.0	49.0	60.0	69.4	90.0	70.1	90.0
575/3/60	No	31.5	40.0	36.7	45.0	55.3	70.0	60.1	80.0
575/3/60	Yes	32.2	40.0	37.4	50.0	56.0	70.0	60.7	80.0

NOTE: High short circuit current rating available on most models. For voltages not shown, contact STULZ.

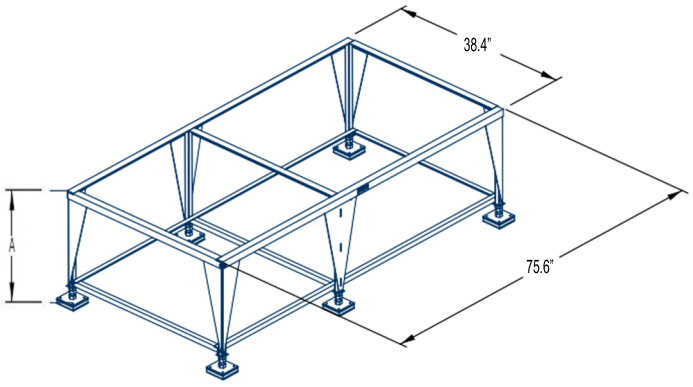
Dimensional Drawings

CFD-021/028/035-D – Downflow

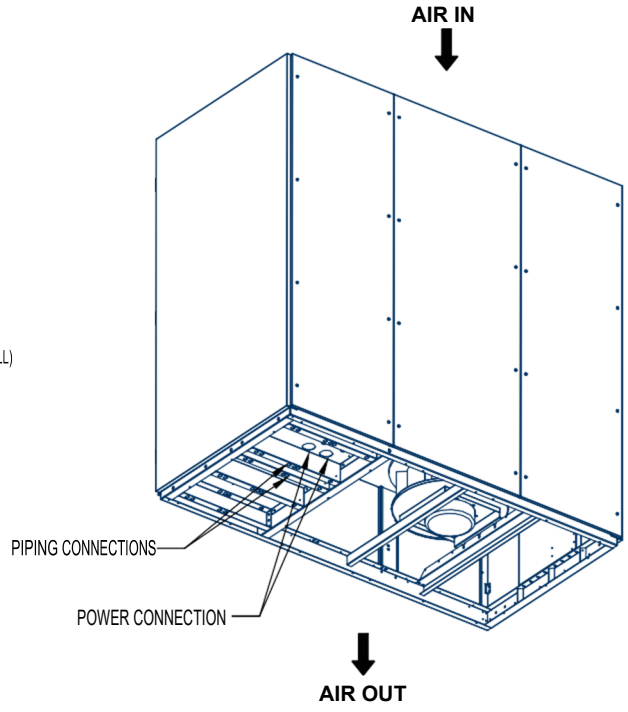
Free Return or with Optional Ducted Return



Adjustable Floor Stand, Non-Seismic



Seismic Rated Floor Stand is Available
Floor Stands Optionally Available



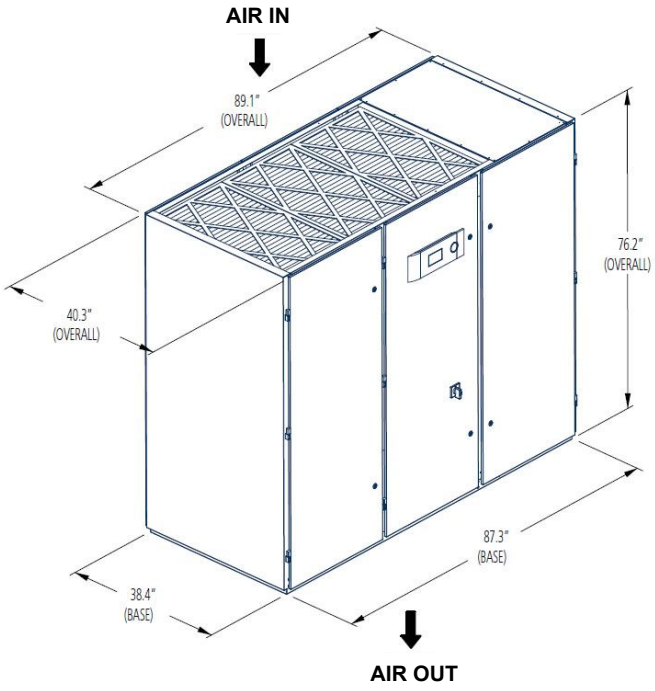
NOTE Standard piping connections terminate inside the cabinet at the bottom, right hand side.

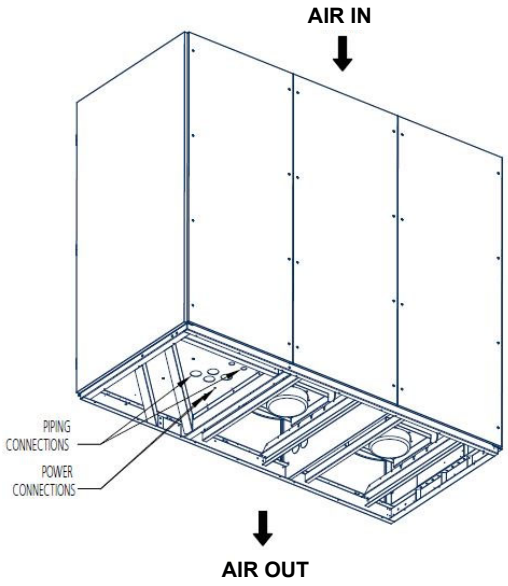
Adjustable Floor Stand Height Dimensions (inches)		
Nominal Height	Dim A Minimum	Dim A Maximum
15.00	14.00	18.00
18.00	17.00	21.00
24.00	23.00	26.00
36.00	35.00	38.00
48.00	47.00	50.00

Dimensional Drawings

CFD-042/053/070-D - Downflow

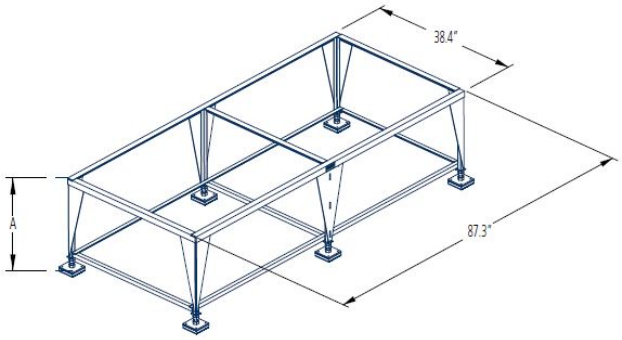
Free Return or with optional Ducted Return





NOTE: Standard piping connections terminate inside the cabinet at the bottom, right hand side.

Adjustable Floor Stand, Non-Seismic
Seismic Rated Floor Stand is Available



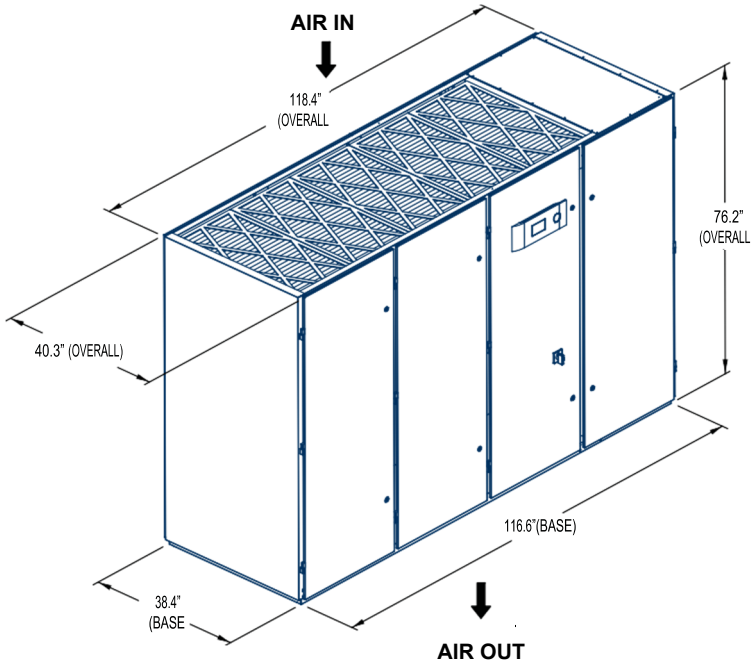
Adjustable Floor Stand Height Dimensions (inches)		
Nominal Height	Dim A Minimum	Dim A Maximum
15.00	14.00	18.00
18.00	17.00	21.00
24.00	23.00	26.00
36.00	35.00	38.00
48.00	47.00	50.00

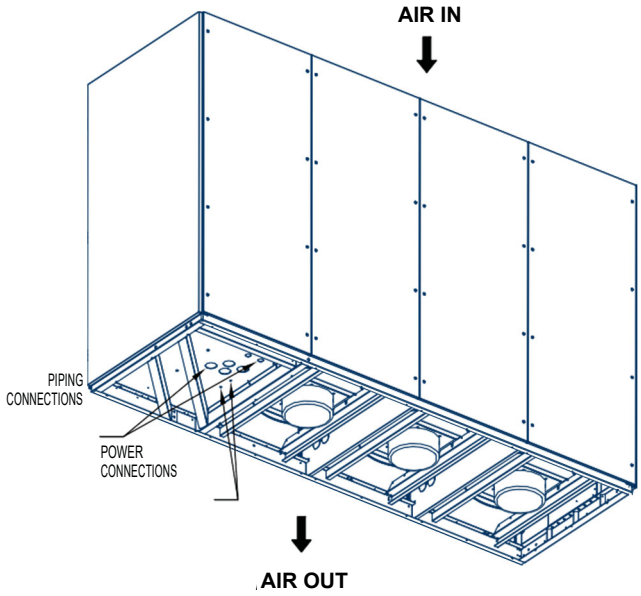
Higher Floor Stands Optionally Available.

Dimensional Drawings

CFD-088/105-D - Downflow

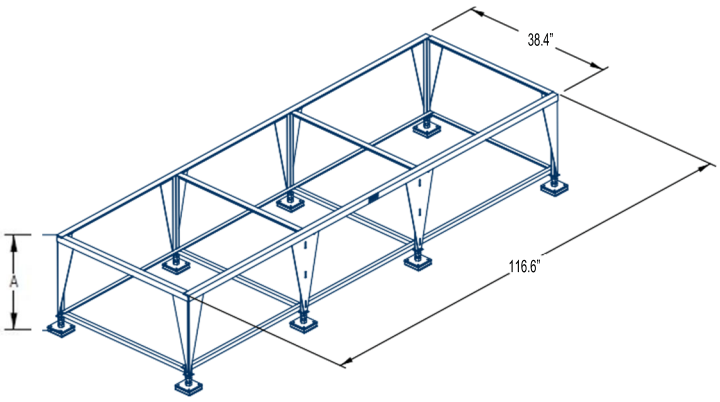
Free Return or with optional Ducted Return





NOTE: Piping connections terminate inside the cabinet at the bottom, right hand side.

Adjustable Floor Stand, Non-Seismic
Seismic Rated Floor Stand is Available



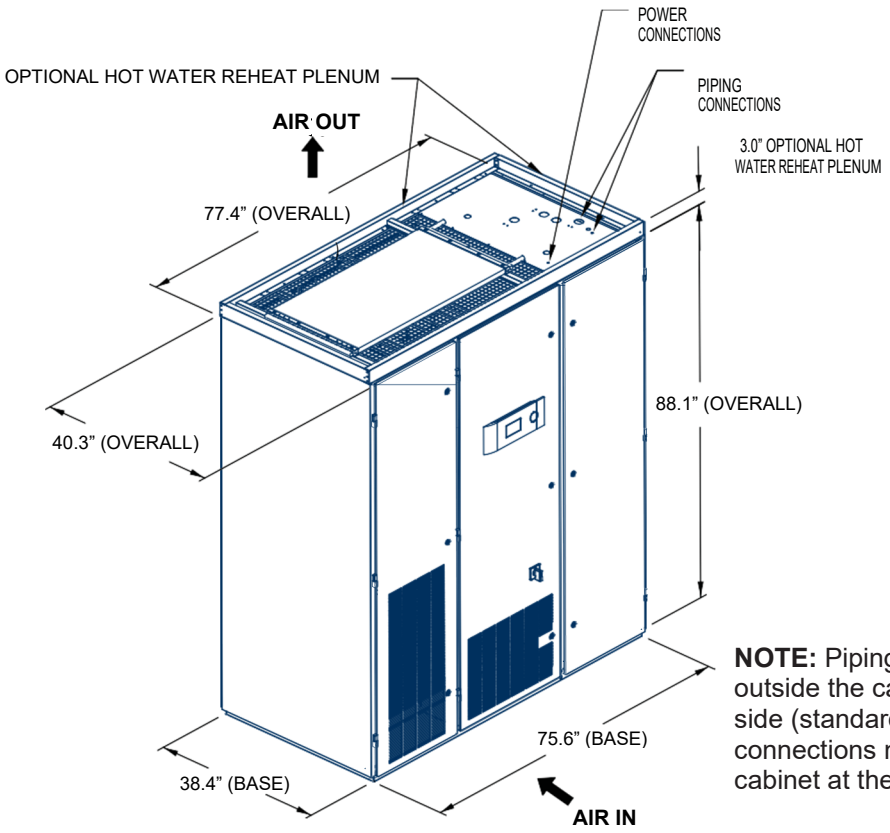
Adjustable Floor Stand Height Dimensions (inches)		
Nominal Height	Dim A Minimum	Dim A Maximum
15.00	14.00	18.00
18.00	17.00	21.00
24.00	23.00	26.00
36.00	35.00	38.00
48.00	47.00	50.00

Higher Floor Stands Optionally Available.

Dimensional Drawings

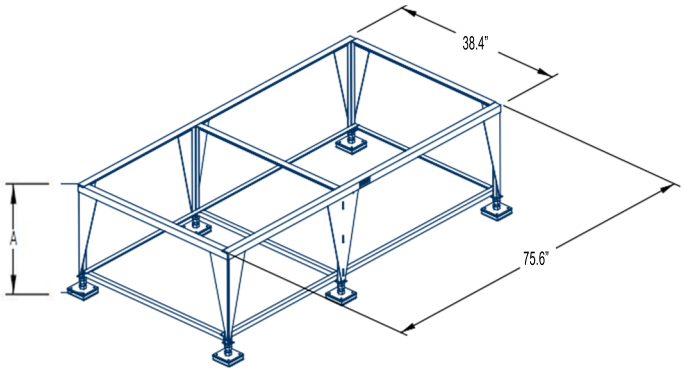
CFU-021/028/035-D – Upflow

Ducted or Provided with Optional 2- or 3-way Plenum Box



NOTE: Piping connections terminate outside the cabinet at the top, right hand side (standard). As an option, piping connections may be furnished inside the cabinet at the bottom right hand side.

Adjustable Floor Stand, Non-Seismic
Seismic Rated Floor Stand is Available



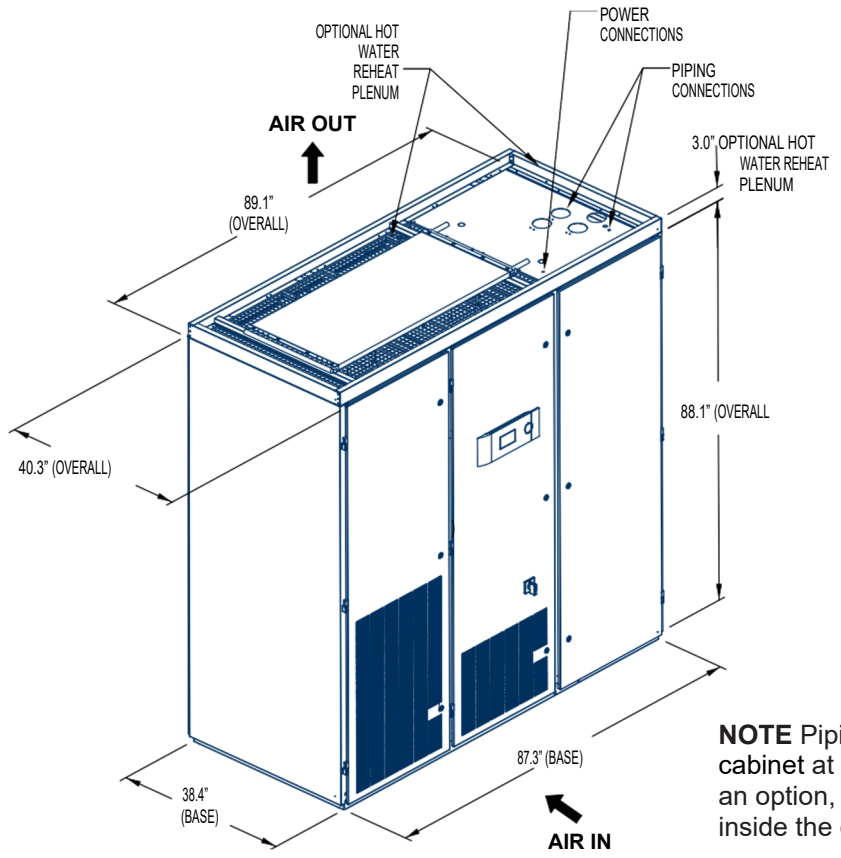
Adjustable Floor Stand Height Dimensions (inches)		
Nominal Height	Dim A Minimum	Dim A Maximum
15.00	14.00	18.00
18.00	17.00	21.00
24.00	23.00	26.00
36.00	35.00	38.00
48.00	47.00	50.00

Higher Floor Stands Optionally Available.

Dimensional Drawings

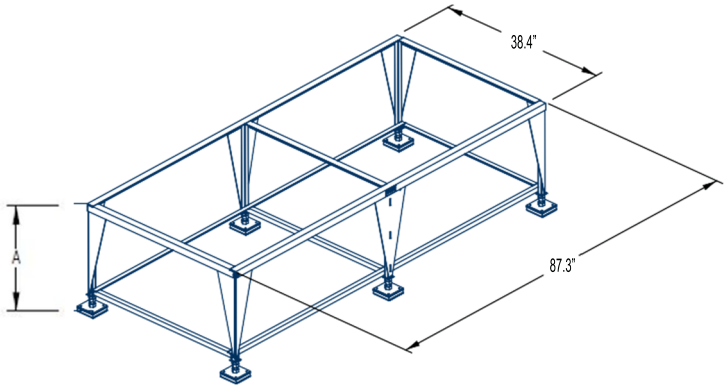
CFU-042/053/070-D – Upflow

Ducted or Provided with Optional 2 or 3-Way Plenum Box



NOTE Piping connections terminate outside the cabinet at the top, right hand side (standard). As an option, piping connections may be furnished inside the cabinet at the bottom right hand side.

Adjustable Floor Stand, Non-Seismic
Seismic Rated Floor Stand is Available



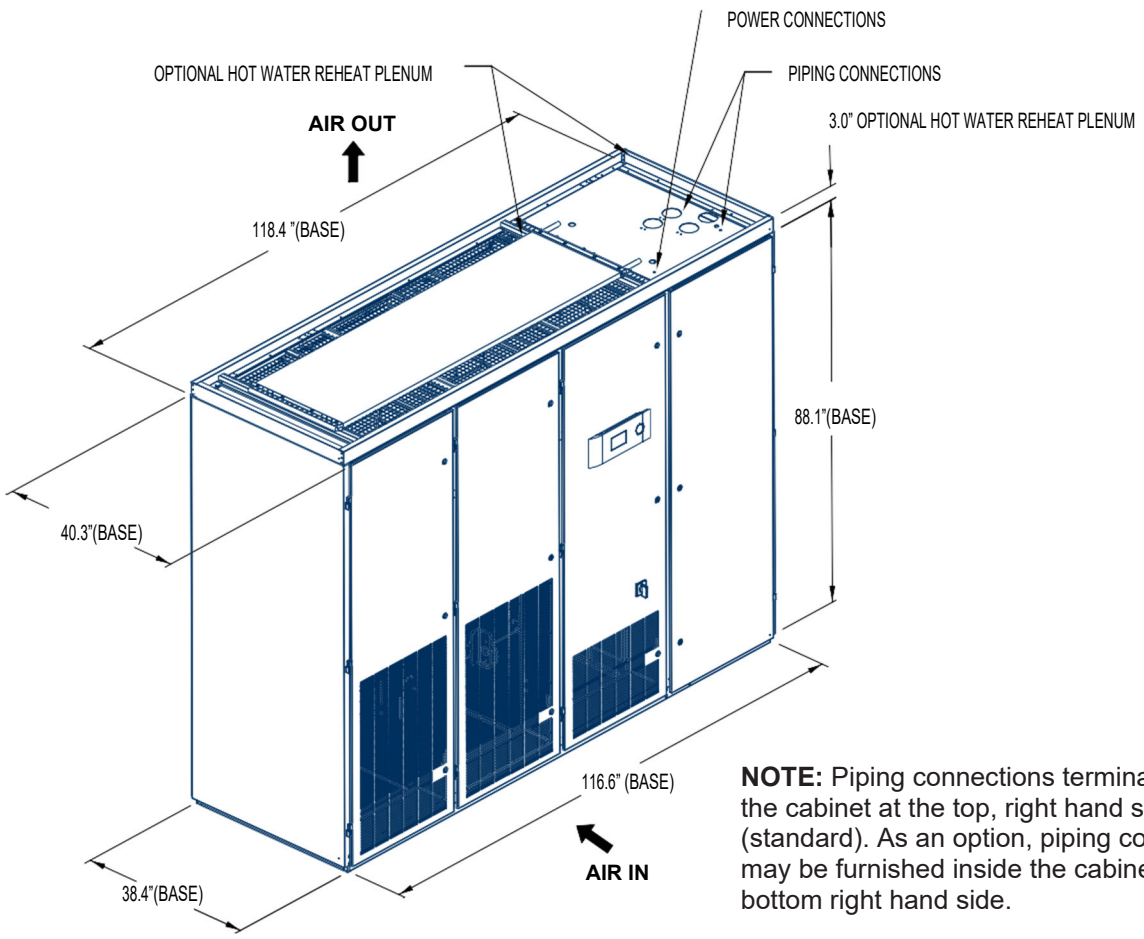
Adjustable Floor Stand Height Dimensions (inches)		
Nominal Height	Dim A Minimum	Dim A Maximum
15.00	14.00	18.00
18.00	17.00	21.00
24.00	23.00	26.00
36.00	35.00	38.00
48.00	47.00	50.00

Higher Floor Stands Optionally Available.

Dimensional Drawings

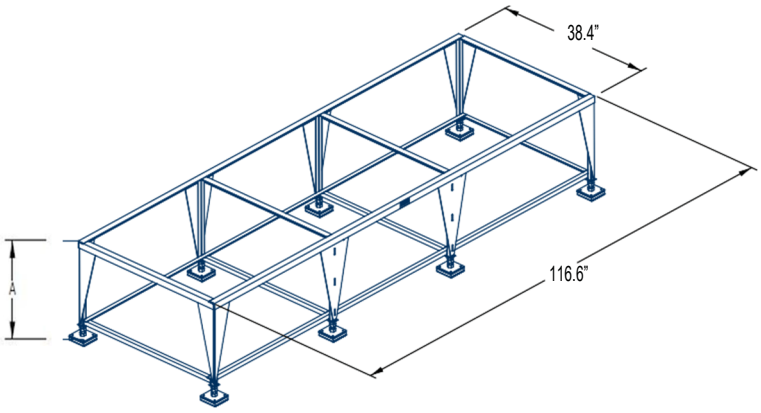
CFU-088/105-D – Upflow

Ducted or Provided with Optional 2 or 3-Way Plenum Box



NOTE: Piping connections terminate outside the cabinet at the top, right hand side (standard). As an option, piping connections may be furnished inside the cabinet at the bottom right hand side.

Adjustable Floor Stand, Non-Seismic
Seismic Rated Floor Stand is Available

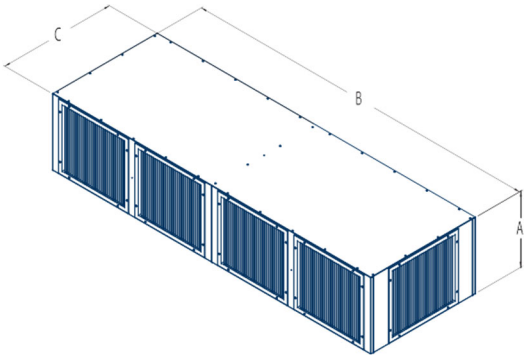


Adjustable Floor Stand Height Dimensions (inches)		
Nominal Height	**Dim A Minimum**	**Dim A Maximum**
15.00	14.00	18.00
18.00	17.00	21.00
24.00	23.00	26.00
36.00	35.00	38.00
48.00	47.00	50.00

Higher Floor Stands Optionally Available.

Supply Air Plenum and Return Air Plenum Extension — Upflow

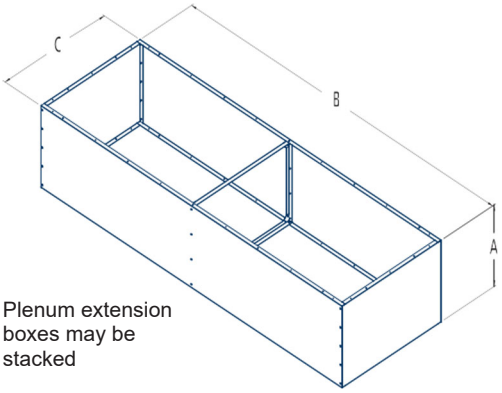
Supply Air Plenums



Plenum Boxes (Up-Flow Units)	Dim A Nominal (inches)	Dim B Minimum (inches)	Dim C Maximum (inches)
CFU-021	18.5	77.4	38.7
CFU-028	18.5	77.4	38.7
CFU-035	18.5	77.4	38.7
CFU-042	18.5	89.1	38.7
CFU-053	18.5	89.1	38.7
CFU-070	18.5	89.1	38.7
CFU-088	18.5	118.4	38.7
CFU-105	18.5	118.4	38.7

Available for Any Size Cabinet

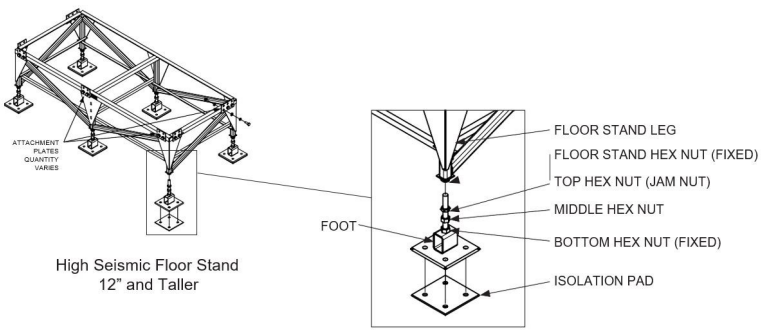
Supply Air Plenum Extension



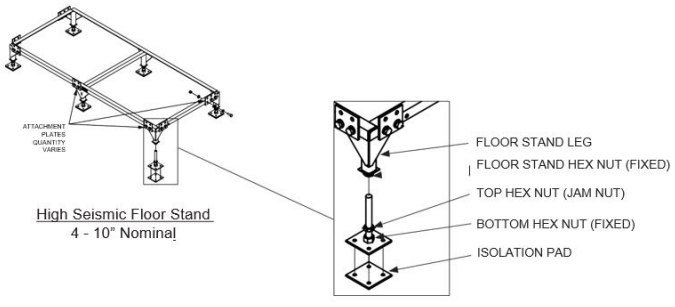
Plenum extension boxes may be stacked

Plenum Extension Boxes (Down-Flow Units)	Dim A Nominal (inches)	Dim B Minimum (inches)	Dim C Maximum (inches)
CFU/CFD-021	24, 36 or 48	77.4	38.8
CFU/CFD-028	24, 36 or 48	77.4	38.8
CFU/CFD-035	24, 36 or 48	77.4	38.8
CFU/CFD-042	24, 36 or 48	89.1	38.8
CFU/CFD-053	24, 36 or 48	89.1	38.8
CFU/CFD-070	24, 36 or 48	89.1	38.8
CFU/CFD-088	24, 36 or 48	118.4	38.8
CFU/CFD-105	24, 36 or 48	118.4	38.8

Typical seismic rated floor stands are below. Enclosed, non-seismic rated floor stands also available.



High Seismic Floor Stand
12" and Taller



High Seismic Floor Stand
4 - 10" Nominal

Guide Specifications

CyberAir® DX

Summary

This specification describes requirements for a precision environmental control system. The CyberAir floor-mounted air conditioning system shall provide precision temperature and/or humidity control for computer rooms or rooms containing communications or other highly sensitive heat load equipment where continuous 24 hour a day, 365 days a year air conditioning is required.

Designed for front service access, CyberAir systems require minimal floor space. The units are designed with a wide range of options to handle all precision cooling applications.

Design Requirements

The environmental control system shall be a CyberAir factory-assembled unit. The unit shall be designed for corner installation requiring front access through hinged and removable front access panels. No allowance for side service access shall be required. CyberAir units are especially adapted for both raised and non-raised floors.

Quality Assurance

The manufacturer shall maintain a set of international standards of quality management to ensure product quality. Prior to shipment each system shall be subject to a complete operational and functional testing based on predefined procedures. The air conditioner manufacturer shall be ISO 9001:2015 certified.

Cabinet

Downflow

Access panels shall be fabricated from 18 gauge galvannealed steel. Door jambs and top cabinet frame shall be fabricated from 16 gauge galvannealed steel. Bottom cabinet frame shall be fabricated from 10 gauge galvannealed steel.

The panels shall be lined with ½" (13 mm), 2 lb (.90 kg), high-density sound and thermal insulation and sealed with a self-extinguishing gasket conforming to NFPA 90A and 90B. The main unit color shall be black, extra fine texture.

Upflow

Access panels shall be fabricated from 18 gauge galvannealed steel. Door jambs and top cabinet frame shall be fabricated from 16 gauge galvannealed steel. Bottom cabinet frame shall be fabricated from 10 gauge galvannealed steel.

The panels shall be lined with ½” (13 mm), 2 lb (.90 kg), high-density sound and thermal insulation and sealed with a self-extinguishing gasket conforming to NFPA 90A and 90B. The main unit color shall be black, extra fine texture.

Air Flow Patterns

Downflow (CFD)

The air conditioner shall be configured for a downflow air pattern with top free return air and conditioned supply air discharge through the bottom of the system into the raised floor.

Upflow (CFU)

The air conditioner shall be configured for an upflow air pattern with free return air through front filtered grille or ducted rear return air and conditioned supply air discharge through the top of the unit.

Air Filtration

All units shall be supplied with disposable air filters classified as UL 900 or UL 586. Filters shall be 4” deep (nominal). Filters shall be pleated with a Minimum Efficiency Reporting Value (MERV) of 8. Filters shall be installed in a front accessible, steel holding frame, and accessible through the front of the unit (except for the rear return configuration).
Optional: Filters rated up to MERV 11 shall be available.

Mechanical Components

Backward Inclined Plenum Style Fan with an EC Motor

The blower(s) shall be direct driven, single inlet, backward curved centrifugal with an electronically commutated motor for maintenance free operation. The motor shall include:

- Integrated electronic control board and direct microprocessor control signaling for fan speed control
- Soft-starting capabilities
- RS-485 BUS connection
- Integrated current limitations

Each fan shall be low noise, low vibration manufactured with a fiberglass reinforced polypropylene impeller. Each fan impeller shall be dynamically and statically balanced in two planes to minimize vibration during operation.

Evaporator Coil

The system shall be configured for draw-thru air pattern to provide uniform air distribution over the coil face. The coils shall be a V-Frame design to provide maximum coil surface area and minimum depth to provide a high sensible cooling capacity. The coils shall be configured as dual refrigeration circuits comprising seamless drawn copper tubes, mechanically bonded to

tempered aluminum fins. Coil end plates shall be hot dipped galvanized steel. The evaporator coil shall be mounted in an insulated stainless-steel condensate drain. Refrigerant flow for each circuit shall be controlled by electronic expansion valves.

Refrigeration System

All refrigerant piping shall be refrigerant grade tubing. Each refrigeration circuit shall include, as a minimum a refrigerant drier/strainer sight glass with moisture detector, an electronic expansion valve, an evaporator coil, a compressor, a high-pressure switch with manual reset, and a low pressure switch with automatic reset. All high-pressure joints shall be brazed, and the entire system shall be pressure tested at the factory with dry nitrogen.

Proportional Electronic Expansion Valve

An electronically operated expansion valve shall be installed to precisely control the flow of liquid refrigerant into the evaporator coil while maintaining the desired superheat across a wide range of operating conditions.

Dual Scroll Compressors

The unit shall be provided with two independently circuited, high efficiency, high reliability, and low noise scroll compressors. Each compressor shall be complete with rotolock service valves, charging and service shraeder ports, internal vibration isolation, internal thermal overloads, an internal pressure relief valve, an internal discharge gas vibration eliminator, and external vibration mounting isolation.

Optional: Tandem scroll compressors are also available on STULZ models CFD-053 and larger for additional capacity control and energy savings. Rotolock service valves are not used with the tandem option.

Snap-Acting Hot Gas Bypass (Optional)

The CyberAir floor-mounted air conditioning system shall incorporate a snap acting hot gas bypass system to provide modulation of the unit’s cooling capacity and evaporator coil freeze protection under low load conditions.

Air Cooled Heat Rejection

-20°F Variable Fan Speed Control

The air-cooled system shall incorporate a low ambient, variable speed fan, head pressure control. The pressure control shall be for year-round air conditioning system operation down to -20°F DB minimum ambient air temperature.

-30°F Flooded Control

The air-cooled system shall incorporate a low ambient, flooded head, pressure control. The pressure control shall be for year-round system operation down to -30°F DB minimum ambient air temperature. Liquid refrigerant receivers, with receiver liquid-level sight glass and head pressure regulator valves (for flooded condenser operation) shall be included. The receiver and heat pressure control valve ships separately as field installed.

Water/Glycol Cooled Heat Rejection

Each evaporator refrigerant circuit shall be provided with a factory installed single pass, counterflow, micro plate heat exchanger. The heat exchanger shall be constructed of type 316 stainless steel; designed and tested for a 650 psig w.w.p.

2-Way, 600 psig Regulating Valve (Standard)

Each refrigerant circuit head pressure shall be controlled by a factory installed 2-way water/glycol regulating valve rated for 600 psig w.w.p. The 2-way condenser water modulating valve shall automatically meter the flow of water to the condenser. It shall do so, in response to a proportional signal (0-10 VDC) provided to the valve by the microprocessor controller.

3-Way, 600 psig Regulating Valve (Optional)

The refrigerant circuit head pressure shall be controlled by a factory installed 3-way water/glycol regulating valve rated for 600 psig w.w.p. The 3-way condenser water modulating valve shall automatically meter the flow of water/glycol to the condenser. It shall do so in response to a proportional signal (0-10 VDC) provided to the valve by the microprocessor controller.

Steam Generating Humidifier (Optional)

The humidifier shall be a self-contained atmospheric steam generator. The humidifier assembly shall include an integral fill cup, fill and drain valves, disposable steam cylinder and associated piping. The humidifier shall be equipped with an auto adaptive control system to optimize water conductivity, control automatic drain/ flush cycles, minimize energy waste and maximize cylinder life.

The humidifier shall have a modulating output between 20% and 100% of rated capacity. The unit shall include draw in water tempering to ensure the drain water do not exceed 140 °F during automatic drain cycles.

Dehumidification Cycle (Standard)

The system shall be provided with a refrigeration-based dehumidification mode. Moisture is condensed on the cooling coil and discharged through the condensate drain. Reheat (electric, hot gas, or hot water) shall be provided to offset sensible cooling of one cooling stage during the dehumidification cycle.

Electric Heat/Reheat

A factory mounted and wired low-watt density, plated fin tubular design electric resistance heater shall be included to provide automatic sensible re-heating as required during the dehumidification cycle and automatic heating mode.

Electric heaters shall be provided with miniature thermal/magnetic circuit breakers, which shall protect each ungrounded conductor. Also included will be one automatic reset and one manual reset over-temperature safety device (pilot duty).

SCR Fired Reheat/Heat (Optional)

The electric heat/reheat shall be controlled through a zero crossing silicon controlled rectifier (SCR) with an extruded aluminum heat sink and solid-state logic system to provide close dry bulb temperature control.

Hot Gas Reheat (Optional)

One cooling stage shall include a factory-installed copper tube, aluminum-fin hot gas reheat coil and valve shall be provided for automatic sensible reheating mode during dehumidification cycle is provided. Hot compressor discharge gas shall be diverted from the condenser to the hot gas reheat coil providing energy-free sensible reheating.

Hot Water Reheat (Optional)

A factory-installed, copper tube, aluminum fin heat/reheat coil and 2-way control valve shall be provided to control the flow of hot water for automatic sensible reheating mode during the dehumidification cycle and automatic heating mode as required.

Electrical System

The electrical system shall conform to National Electrical Code requirements. The control circuit shall be 24 volts AC, wired in accordance with NEC Class II requirements. The control circuit wire shall not be smaller than 18 AWG. All wiring shall be neatly wrapped and routed in bundles. Each wire shall end with a service loop and be securely fastened by an approved method. Each wire in the unit shall be numbered for ease of service tracing.
All electrically actuated components shall be easily accessible from the front of the unit without reaching over exposed high voltage components or rotating parts. Each high voltage circuit shall be individually protected by circuit breakers or manual motor starters on all three phases. Line voltage and 24-volt control circuit wiring shall be routed in separate bundles.

The electric box shall be positioned for service convenience and shall include all the contactors, starters, fuses, circuit breakers, terminal boards and control transformer required for operation of the unit and shall allow for full service access.

Main Power Service Switch

The unit shall be provided with a unit mounted main power service non-fused disconnect switch.

Remote Stop/Start Contacts

Included in the system's electrical control circuit shall be a 2-pin terminal connection for remote stop/start of the CyberAiR CW air conditioner by remote source.

Automatic Dual Power Transfer Switch (Optional)

Two individual main power input disconnect switches shall be provided, one for each incoming power source. In the event the primary power fails, or a phase loss/imbalance occurs, the automatic transfer switch immediately transfers power to the secondary power source. Once the primary power has been restored, the transfer switch automatically shifts the power load back to the primary power. The transfer time from one source to the other is adjustable to allow staging or to sequence restart of load.

In addition to the automatic transfer switch, the local controller display shall indicate which power source has failed. This indication is a visual depiction that allows the user to determine the status of the input source. This status shall be conveyed though the BMS serial communications link. A through-the-door indicator light shall be included for a fast and local notification of available power sources

Airflow Control

EC Fan Speed Control

The system shall include a variable fan speed control package. The purpose of this feature is to allow the use of the energy saving capabilities of the EC fan without running into the problems associated with ice formation on the coil or excessive latent cooling. When the suction pressure and therefore suction temperature is equal to the minimum allowable coil temperature, the E² controller will prevent the EC fans from any further decrease in speed until that suction pressure/temperature rises. The minimum allowable suction temperature is a customer selectable parameter.

Fail Safe Mode

In the event of a fan failure, the controller shall automatically enable all remaining EC fans to increase to the maximum fan speed setting.

E² Series Controller

The advanced microprocessor based E² Series controller shall be equipped with flexible software capable of meeting the specific needs of the application. The setpoints shall be default

and their ranges shall be easily viewed and adjusted from the user interface display. The program and operating parameters shall be permanently stored on a non-volatile system in the event of power failure. The controller shall be designed to manage temperature and relative humidity (RH) levels to a user defined setpoint via control output signals to the CyberAiR system. Control parameters have variable outputs from 0 to 100% of the full rated capacity. The controller shall receive inputs for measurable control conditions (temperature, relative humidity, and dew point) via return air or room mounted sensors. The internal logic will then determine if the conditions require cooling, humidification or dehumidification. Control setpoints shall be established to maintain design conditions of the installation. The controller will respond accordingly to changes in these conditions and control the output/demand for the appropriate mode of operation until user defined conditions are achieved.

Large Bezel Display Panel — Touch Screen

The large bezel touch screen user interface display panel features a high-resolution backlit liquid-crystal graphical display equipped with contrast adjustment and LED illuminated function keys. The screens that appear on the user interface display panel present data that is from the controller.

The controller offers an alarm log plus four different interface menu levels to the operator: Information, Control, Service, and Factory. These menus permit the user to easily view, control, and configure operating parameters for the CyberAir system.

Field Configurable

The program for the E² Series controller shall be field configurable, allowing the operator the capability of selecting control setpoints specific to the application.

Password Protection

Access to the Info Menu, Alarms Log, and the ability to monitor room conditions shall be allowed without the use of a password. Modifications to the control setpoints shall require the use of a password. The controller shall be programmed to recognize predetermined security levels before allowing access to display screens containing critical variables.

Three secured menu levels (Control, Service, and Factory) will support unique passwords that must be entered to access the menu screens so only authorized personnel may perform modifications to the settings.

Restorable Parameters/Factory Defaults

Upon initial start-up, the CyberAir system shall operate using the setpoints programmed by the factory. The customer may enter new operating parameters in the Control menu and the system will then operate accordingly. The new setpoints may be stored as Customer Default setpoints. The primary setpoints entered by the factory remain stored in the controller’s memory as, factory setpoints.

The setpoints for the system may be re-adjusted in the Control menu at any time. If it becomes necessary, the customer may restore the setpoints back to the Customer Default setpoint values or to the original Factory (primary) setpoint values.

A/C Grouping pLAN Operation (Optional)

Multiple CyberAir system controllers shall be able to be connected (grouped) to a pLAN local network to enable the communication of data and information from each controller to a central control terminal or Lead controller.
The Lead controller display screen can be used to monitor and adjust group control variables for the individual system controllers. Each E² controller connected to the pLAN network shall be identified with its own unique address.

Multiple CyberAir systems consisting of up to eight floor-mounted air conditioners equipped with like controllers may be controlled and monitored via the E² series controller. With multiple CyberAir systems, each unit can be selectively configured as Active to operate as a primary A/C, Capacity Assist for staged operation, or as Standby to come online in case of a failed air conditioning unit to ensure continuous availability.

The controller may also be configured to rotate units with timed duty cycling to promote equal run- time and assure that each CyberAir system within the rotating group is operationally exercised on a periodic timed basis.

Remote BMS Communications (Optional)

The E² series controller shall incorporate a communication interface port that can be field connected to a Building Management System via Modbus RTU, BACnet MS/TP, SNMP, HTTP, or BACnet over ETHERNET/IP as configured by the factory. A controller interfaced to a network must be configured for BMS communication.

Summary Alarm

Alarm conditions shall activate a red LED indicator that backlights the alarm function key. As an option, an alarm condition may also be enunciated by an audible alarm signal. An alarm is acknowledged by pressing the alarm key.

This calls up alarm display screens that provides a text message detailing the alarm conditions. After an alarm condition is corrected, the alarm can be cleared by pressing the alarm key.

Remote Start/Stop Contacts

Included in the system’s electrical control circuit shall be a 2-pin terminal connection for remote start/stop of the CyberAir CW air conditioner by a remote source.

E² Ultracap Constant Contact (Optional)

The control package shall include a Ultracap Constant Contact short term power supply for the E² Series Controller to allow the controller to manage the power transfer switch from one

primary power source to another while maintaining communications with an active monitoring system. The E² Ultracap Constant Contact shall permit a minimum of one-minute ride thru period.

Optional Alarms

CyberAir floor-mounted air conditioning system standard features can be deleted and/or substituted with optional features to allow you the flexibility to select the configuration best suited for your application.

Adjustable Floor Stand

An adjustable floor stand shall be provided to enable ease of installation of the CyberAir floor-mounted air conditioning system onto a raised floor environment. Floor stand height shall be adjustable and shipped separately for field installation.

Seismic Rated Floor Stand

The unit floor stand is constructed and rated for specific seismic requirements. Please contact your local representative for seismic design details.

Enclosed Floor Stand

A factory provided enclosed floor stand shall be factory constructed full cabinet length and pre-fabricated within the support structure for one-way front air discharge. A factory-provided enclosed floor stand is constructed to fit the full cabinet length, and pre-fabricated within the support structure for one-way, front air discharge.

Condensate Pump

A condensate pump shall be factory installed within the CyberAir floor-mounted air conditioning system for automatic removal of condensate and humidifier flush water (if applicable).
The condensate pump shall include an internal overflow safety float which, when wired to the remote start/stop terminals, shall open the unit's control circuit, thereby shutting the unit down in the event of a condensate overflow.

The condensate pump shall be specifically designed to operate with the higher condensate temperatures caused by the flush and drain cycle of the electrode canister humidifiers.

Smoke Detection

A photo-electric smoke detector shall be factory installed and wired in the return air section of the CyberAir floor mounted air conditioning system. The photo-electric detector shall include built-in circuitry that performs a functional test of all detection circuits at least once every 40 seconds without the need for generating smoke. The UL-listed velocity range shall be 0-3000 fpm. The air conditioner will shut down upon sensing smoke in the return air stream.

Firestat

The CyberAir floor-mounted air conditioning system shall be provided with a factory wired and mounted firestat. The firestat will shut down the air conditioner upon sensing a high return air temperature.

Remote Water Detector— Spot Type

A remote single-point water/leak detector shall be factory supplied and shall ship separately for field installation. Upon sensing a water leak, the normally closed water detector control circuit shall open, thereby shutting down the CyberAir floor-mounted air conditioning unit's water producing components.

Remote Water Detector— Dual Spot Type

A dual remote single point water/leak detector shall be factory supplied and shipped separately for field installation. Upon sensing a water leak, the normally closed water detector control circuit shall open, thereby shutting down the CyberAir floor-mounted air conditioning unit's water producing components.

Remote Water Detector— Strip Type

A 20 ft. remote strip/cable type water/leak detector shall be provided for remote field installation. Upon sensing a water leak, the normally closed water detector control circuit shall open, thereby shutting down the CyberAir floor-mounted air conditioning unit's water producing components.

Supply Air Discharge Plenum

A 2- or 3-way plenum discharge box shall be provided. The plenum box shall include double-deflecting, adjustable grilles. It shall be selectable for upflow units only.

Return Air Plenum Extension

A return air plenum extension shall be provided to raise the return air connection. The plenum extension shall be coated to match the CyberAir air conditioning system.

High Short Circuit Current Rating

The CyberAir floor-mounted air conditioning system shall be rated for a short circuit current rating for a maximum of 65k AIC. The higher short circuit current rating shall include safe touch fusing upstream of the unit's main power disconnect switch.

Free-Cooling Mode (Optional)

DX Water/Glycol Cooled systems with Free-Cooling are provided with the following standard DX head pressure and Free-Cooling valve combination:

- DX Valves – 3-way, 600 psig
- FC Valve – 3-way, 600 psig (021-035 kW Systems) 3-way, 400 psig (042-105 kW Systems)

A Free-Cooling economizer cycle shall be provided to take advantage of low ambient air temperature conditions to provide compressor less cooling. The air conditioner shall have three modes of operation:

- 100% Free-Cooling cycle during low ambient air conditions
- Assist FC + Stage DX Mode to provide partial free-cooling with one DX compressor as an assist mode
- 2-stage DX operation without Free-Cooling active due to higher ambient temperatures

Alternate Water Source (Optional)

This is chilled water by day and DX backup by night.

An alternative water source cooling cycle shall be provided to use building chilled water supply when available as the primary cooling cycle with dual compressor, DX refrigerant cooling as a backup. The air conditioner shall have two cooling systems: The primary code of cooling shall be chilled water/glycol circuit with alternative water source cooling coil and the secondary/backup mode of cooling shall be a dual compressor, DX refrigerant circuit.

The E² microprocessor's Alternative Water Source program algorithm shall analyze input data from factory provided and installed room air temperature and relative humidity sensors. Based on this data, the E² controller shall automatically control the sequencing of the primary AWS chilled water to and from the secondary /backup dual compressor DX modes of operation. If chilled water is available the system will operate like a chilled water unit, without the compressor operating. When the water temperature is too high, or the water flow rate is not sufficient, the air conditioner automatically switches to the dual compressor, DX refrigerant cycle.

Alternate water source cooling shall be controlled by the following standard and optional control valves:

- 2-way, 600 psig (Standard)
- 3-way (Optional) 600 psig Rated (021-035 kW Systems)
- 400 psig Rated (042-105 kW Systems)

Modulating AWS cooling control valve (2-way standard, 3-way optional) shall be factory installed. Precision cooling control shall be accomplished via an analog control signal to the proportionally actuating control valve.

Air-Side Economizer Controls

The CyberAir floor-mounted air conditioning system controller shall be equipped with a unique air-side economizer mode for applications using outdoor air-side economizing.

The Air-Side economizer control package shall include an outdoor temperature and humidity sensor, remote space return air sensor, and supply air sensor for proper control during economizer operation. An individual analog output signal (0-10 VDC) is provided to control the external outdoor air intake damper. Damper provided by others.

Control settings shall be included for both temperature and humidity properties of the outdoor air. A damper signal lockout shall be included if the outdoor air conditions reaches user adjustable limits. The economizing damper signal shall allow a minimum output setting for minimum outside air control to meet ventilation requirements.

Code Conformance

The supplied system shall be provided with certification by a NRTL to the following standards:
CETL US listed to UL 1995 (2011 Ed. 4)
CSA C22.2 No. 236 (2011 Ed. 4)

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